

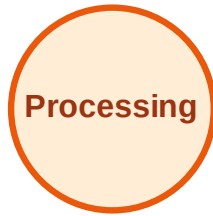
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



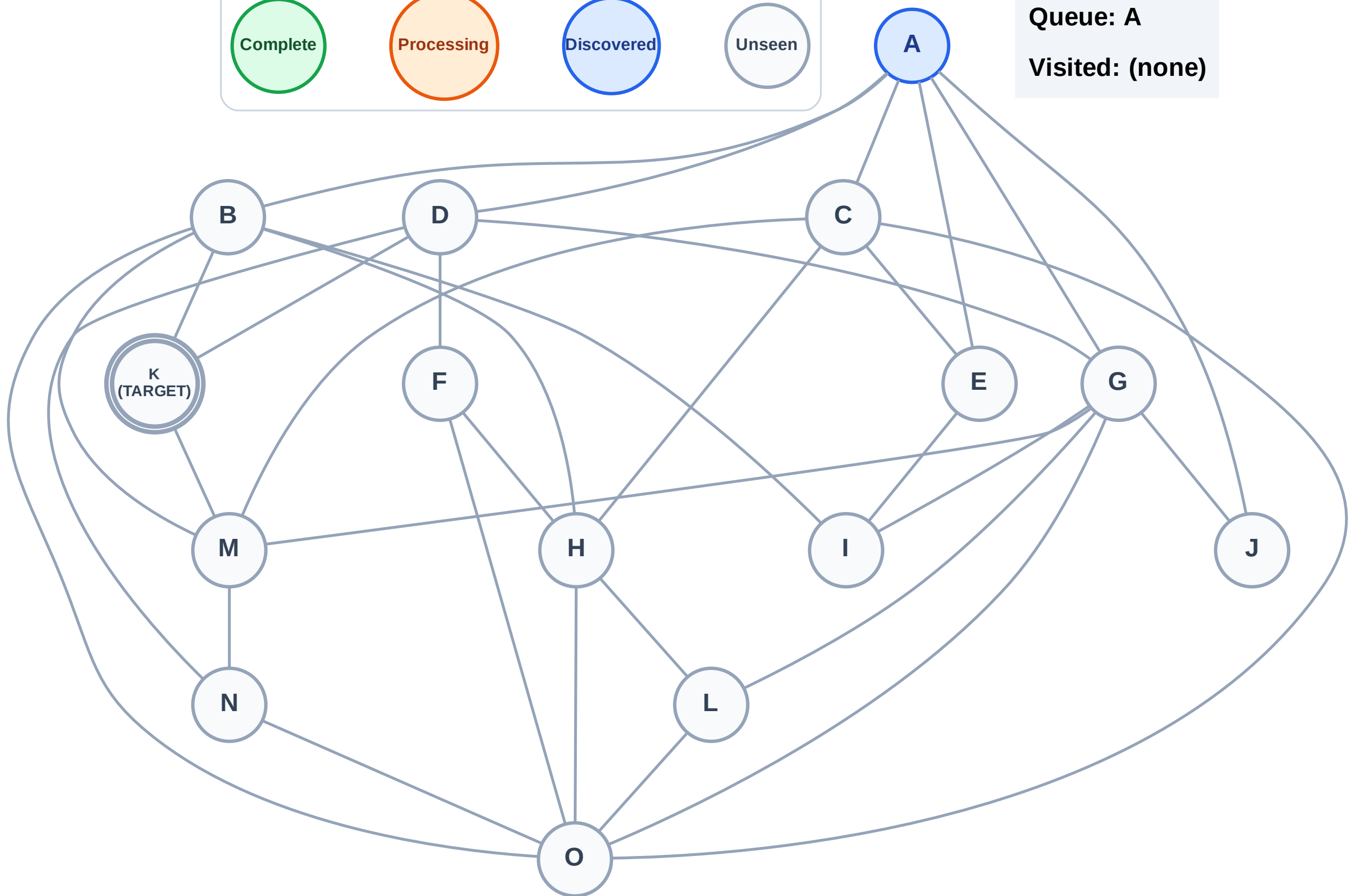
Discovered



Unseen

Queue: A

Visited: (none)



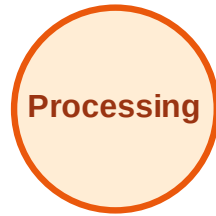
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



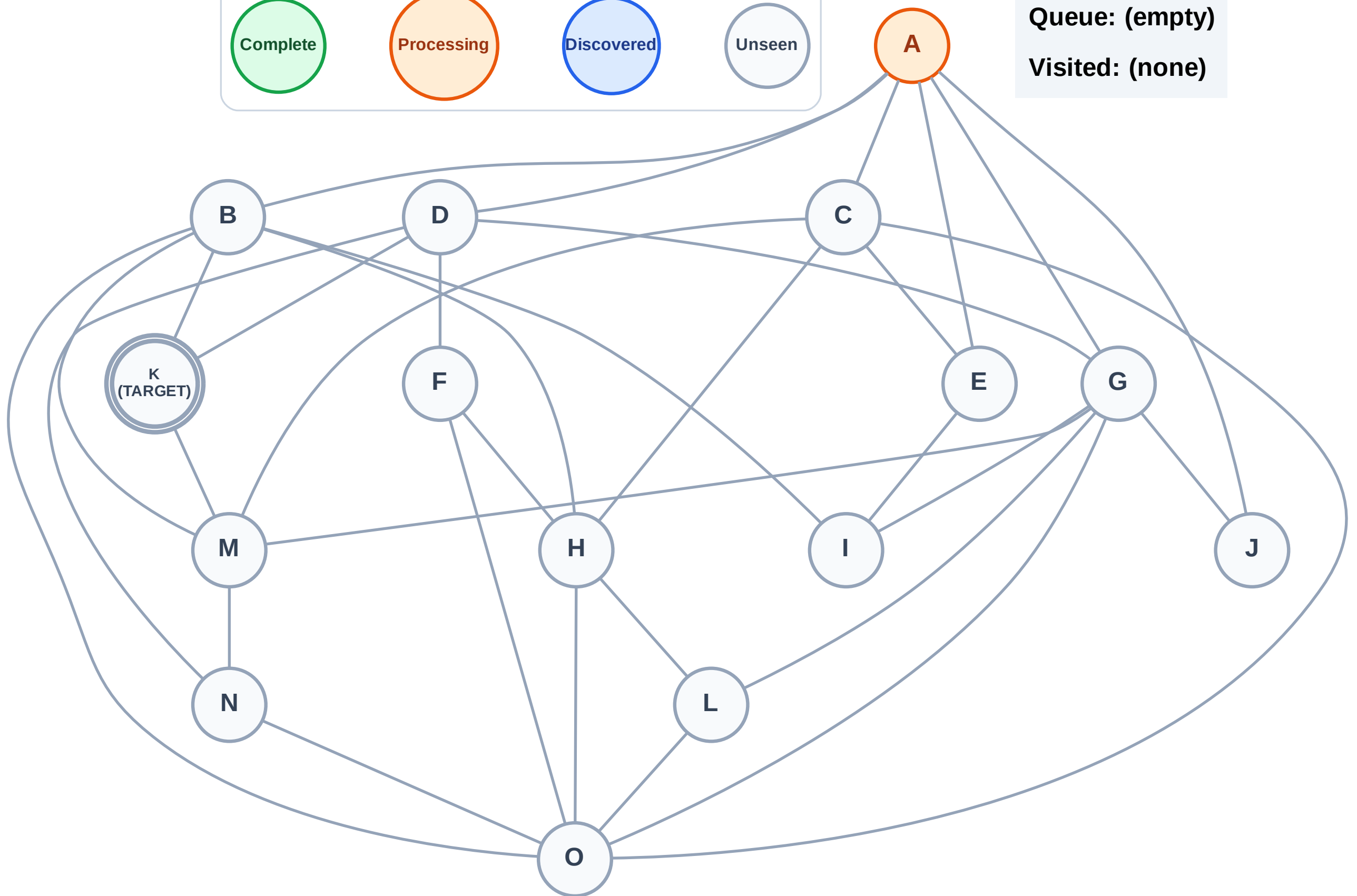
Discovered



Unseen

Queue: (empty)

Visited: (none)



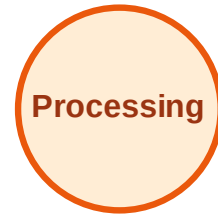
BFS Traversal

Step 3: Discover B, C, D, E, G, J from A

Legend



Complete



Processing



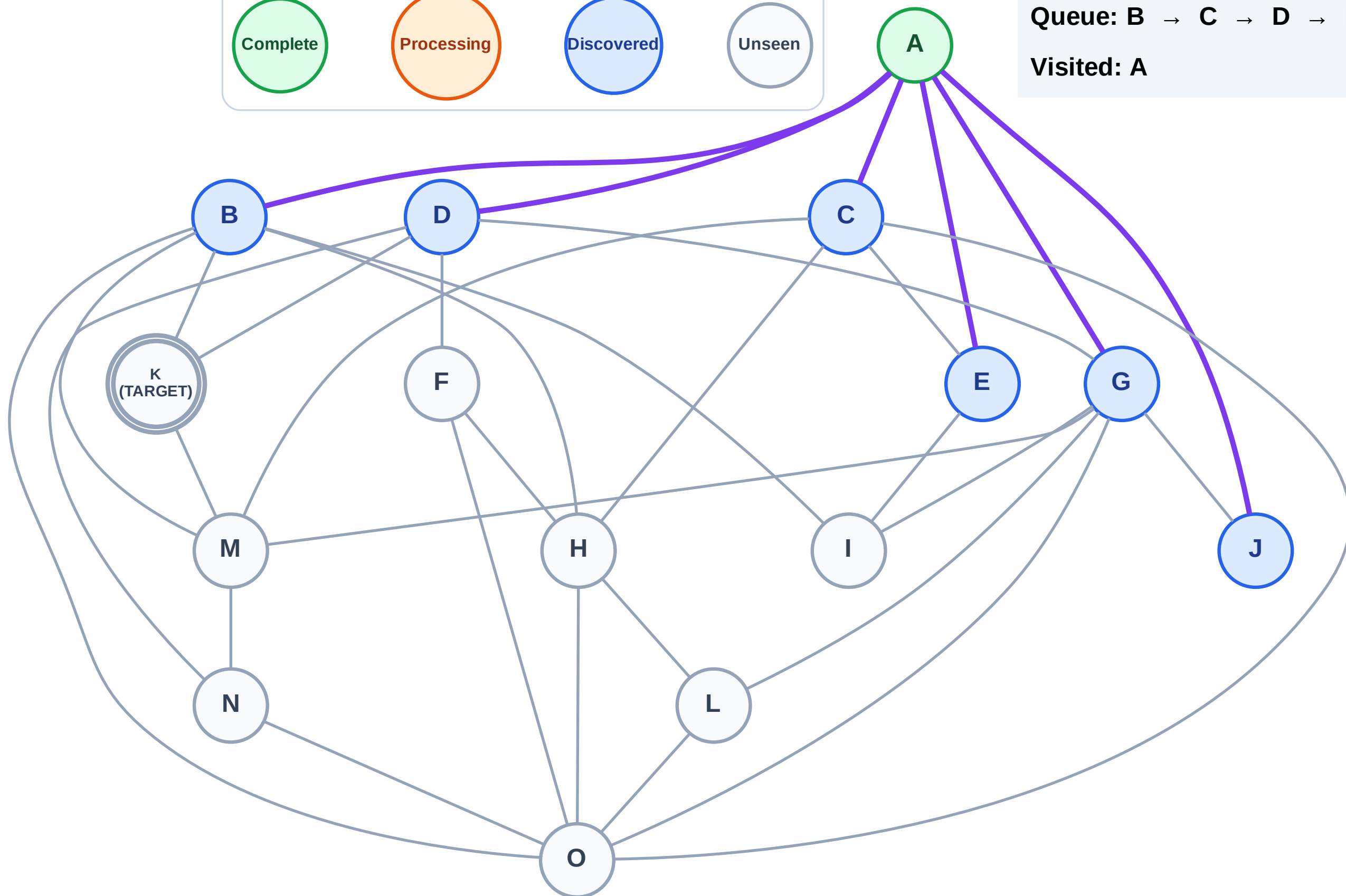
Discovered



Unseen

Queue: B → C → D → E → G → J

Visited: A



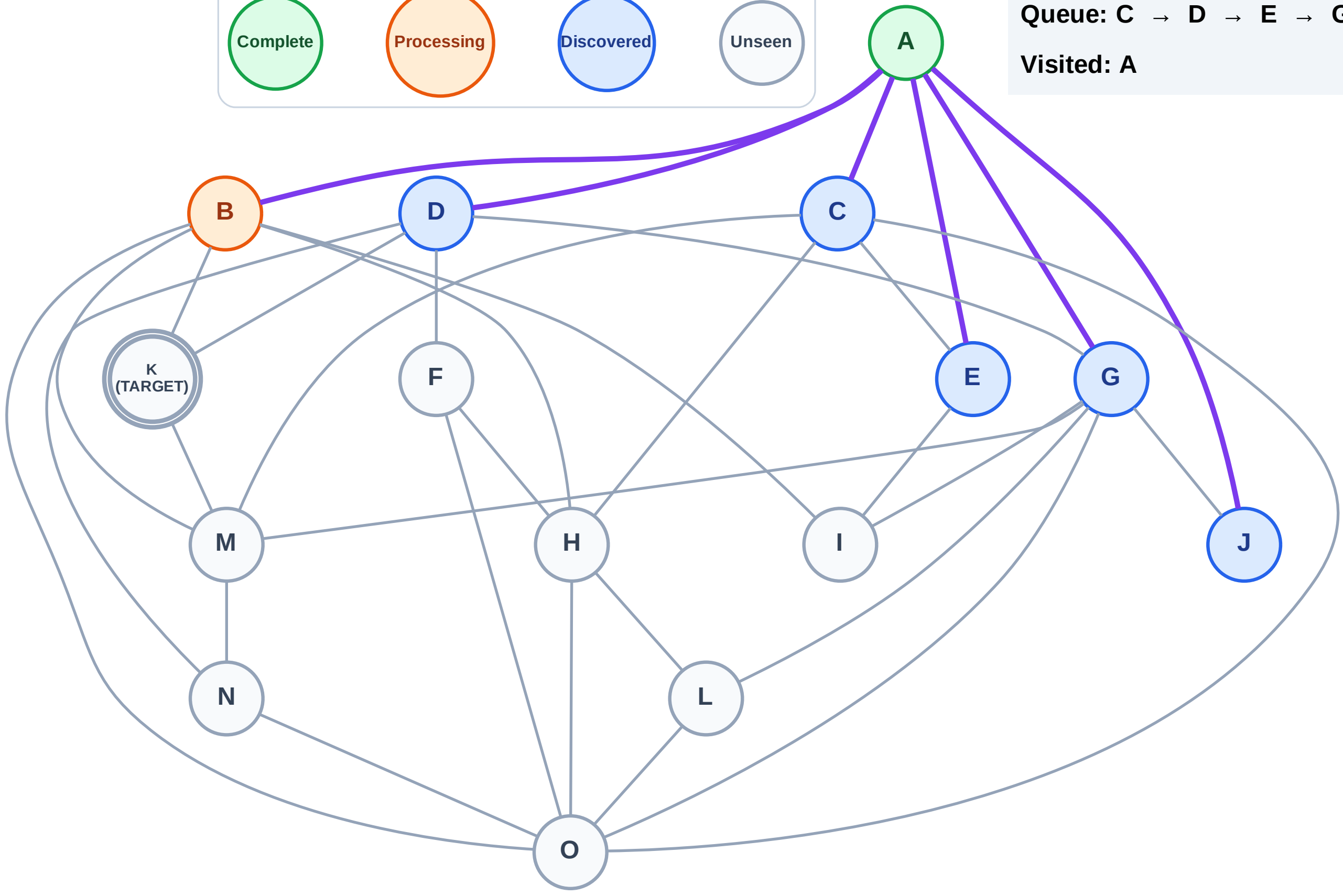
BFS Traversal

Step 4: Process node B

Legend



Queue: C → D → E → G → J
Visited: A



BFS Traversal

Step 5: Discover H, I, K, M, O from B

Legend

Complete

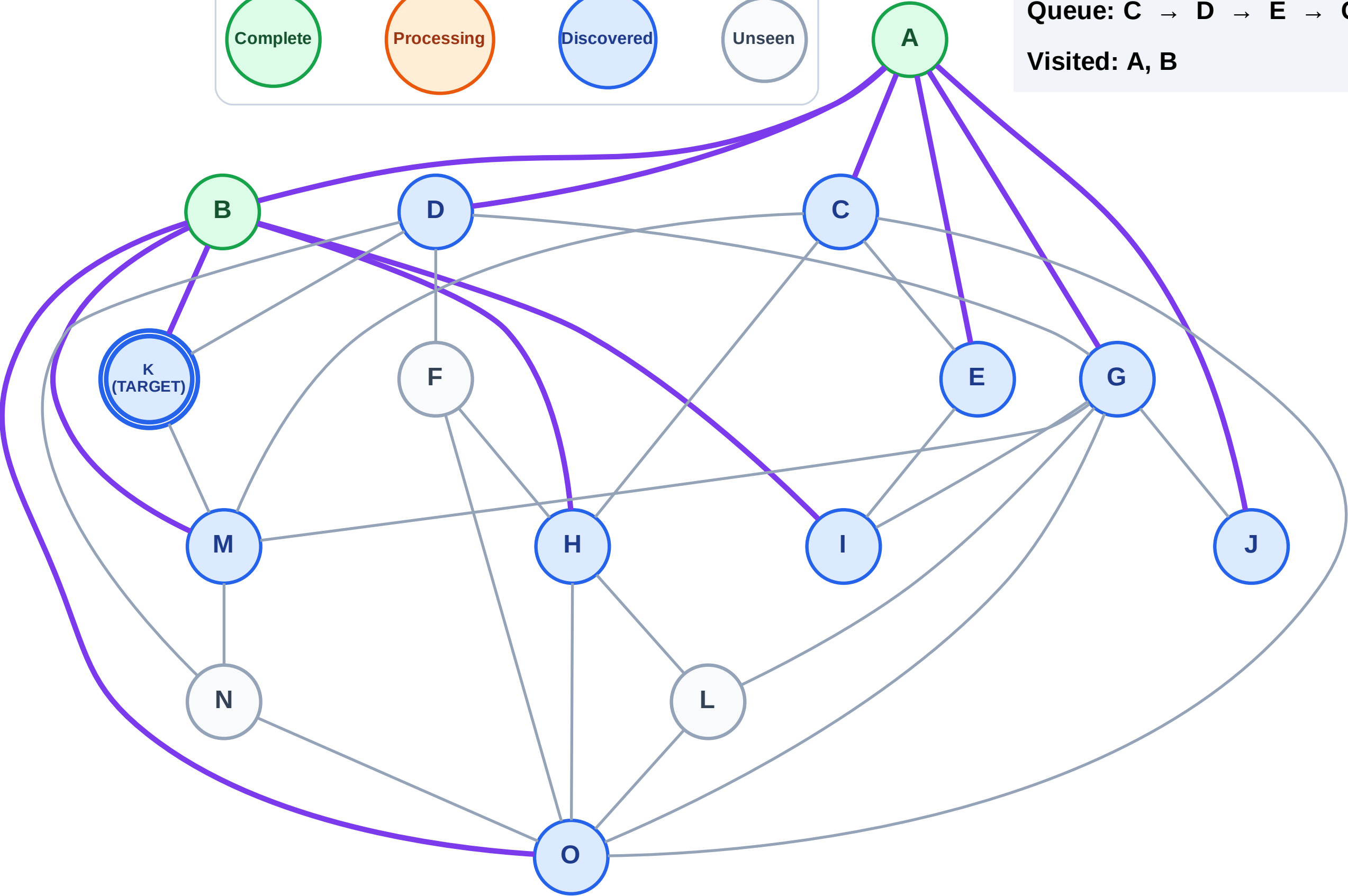
Processing

Discovered

Unseen

Queue: C → D → E → G → J → H → I → K → M → O

Visited: A, B



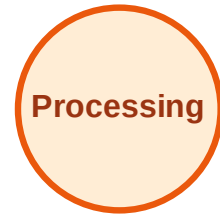
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



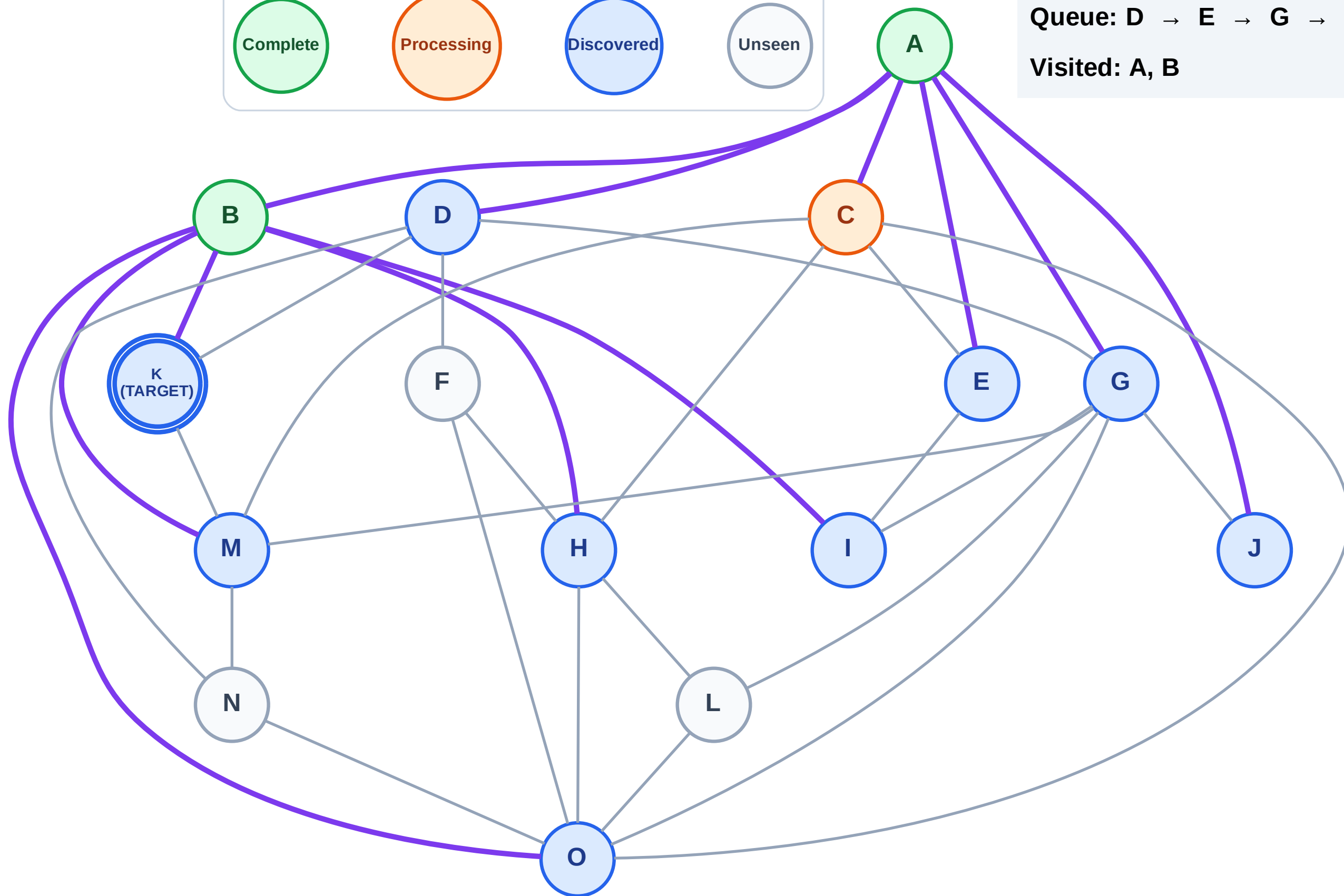
Discovered



Unseen

Queue: D → E → G → J → H → I → K → M → O

Visited: A, B



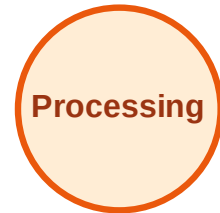
BFS Traversal

Step 7: No new neighbors from C

Legend



Complete



Processing



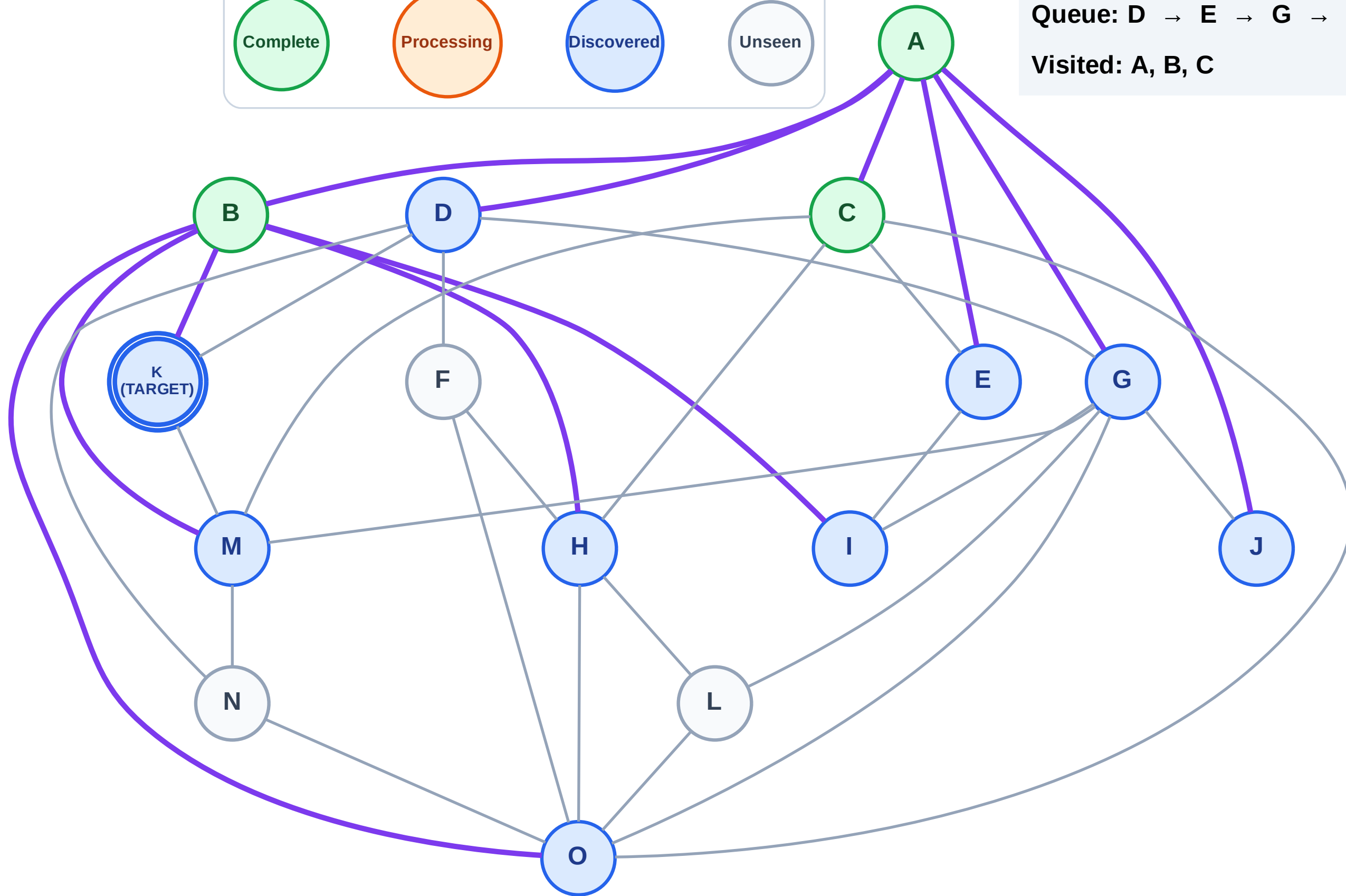
Discovered



Unseen

Queue: D → E → G → J → H → I → K → M → O

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

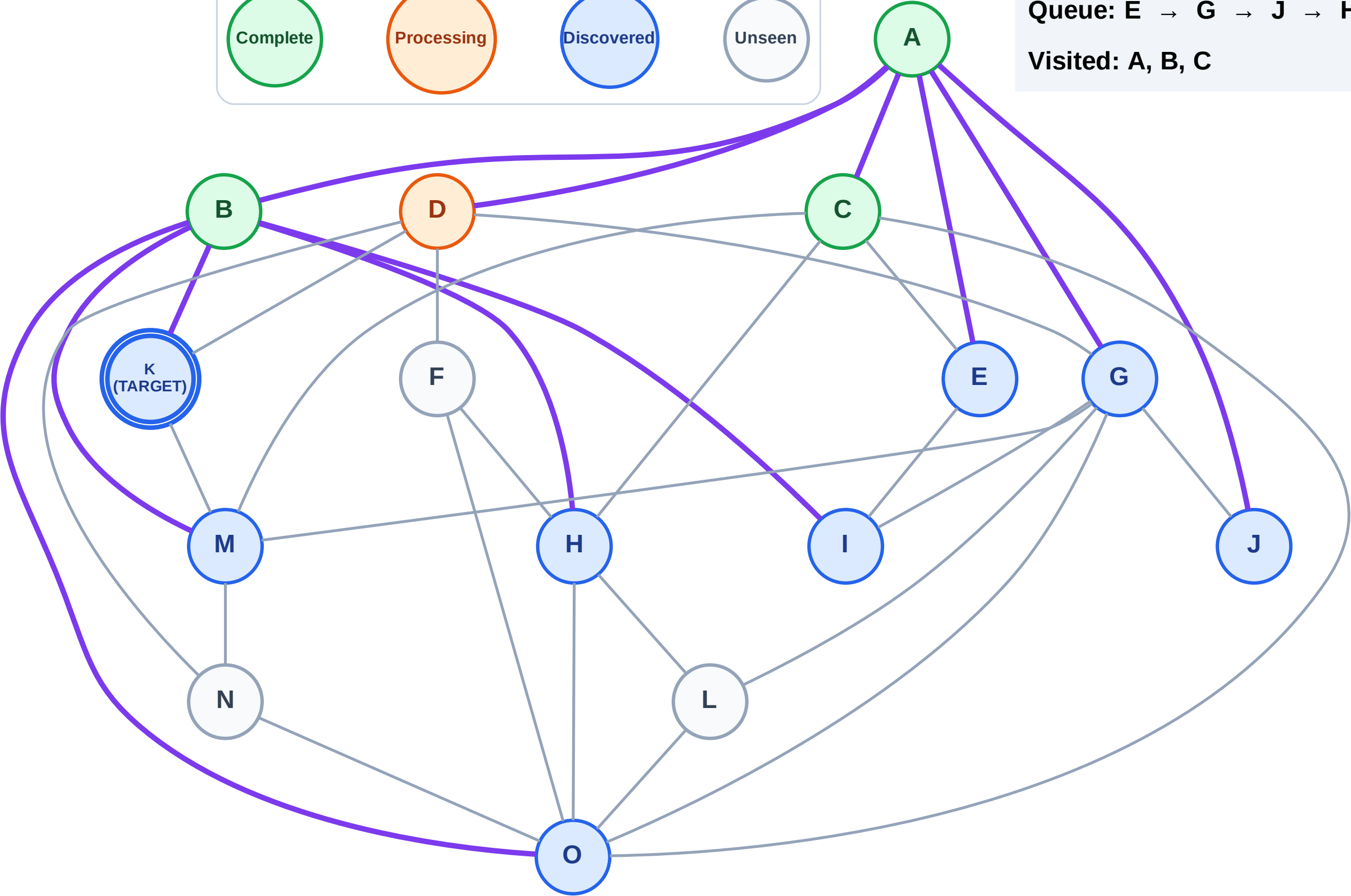
Processing

Discovered

Unseen

Queue: E → G → J → H → I → K → M → O

Visited: A, B, C



BFS Traversal

Step 9: Discover F, N from D

Legend



Complete



Processing



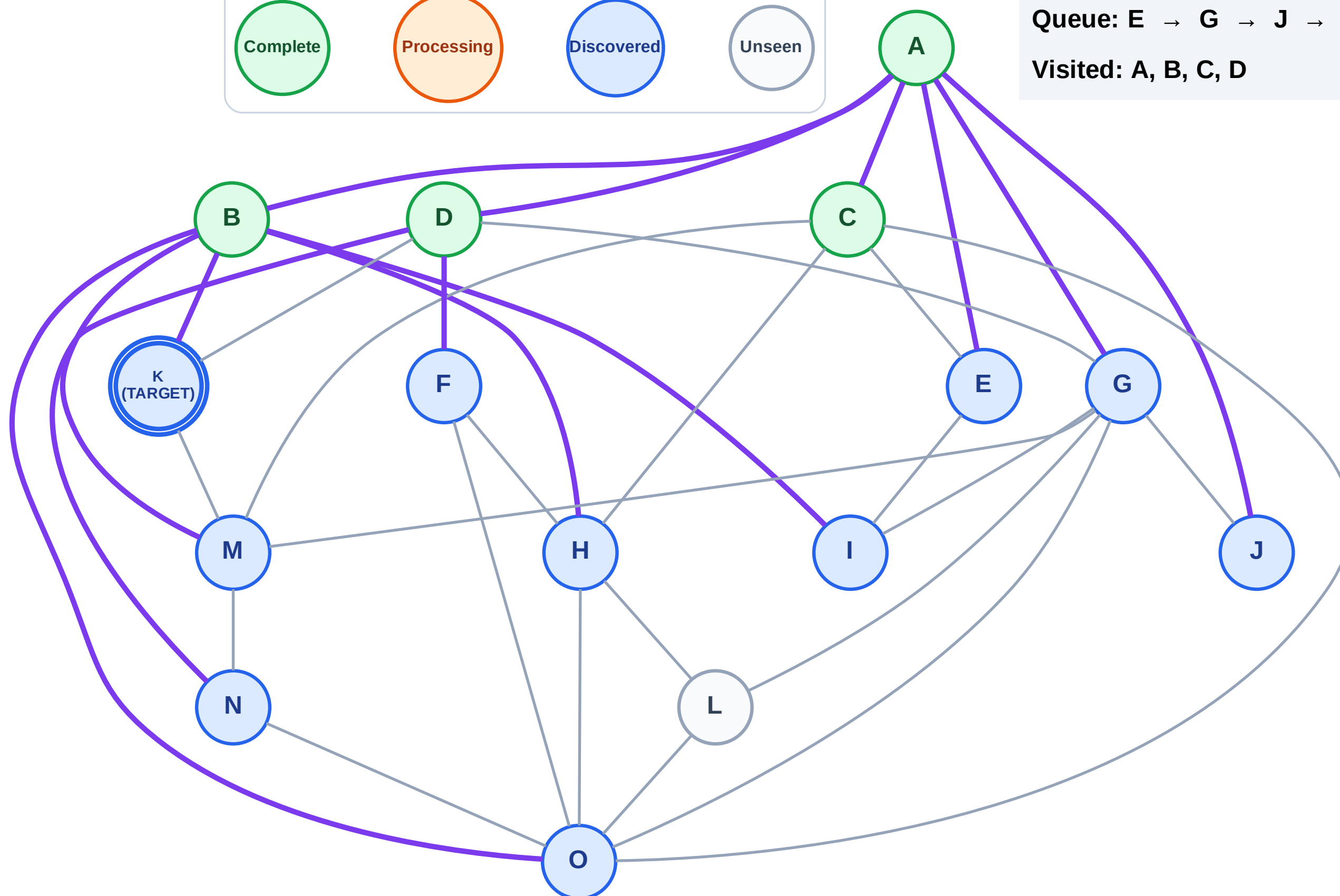
Discovered



Unseen

Queue: E → G → J → H → I → K → M → O → F → N

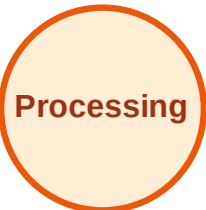
Visited: A, B, C, D



BFS Traversal

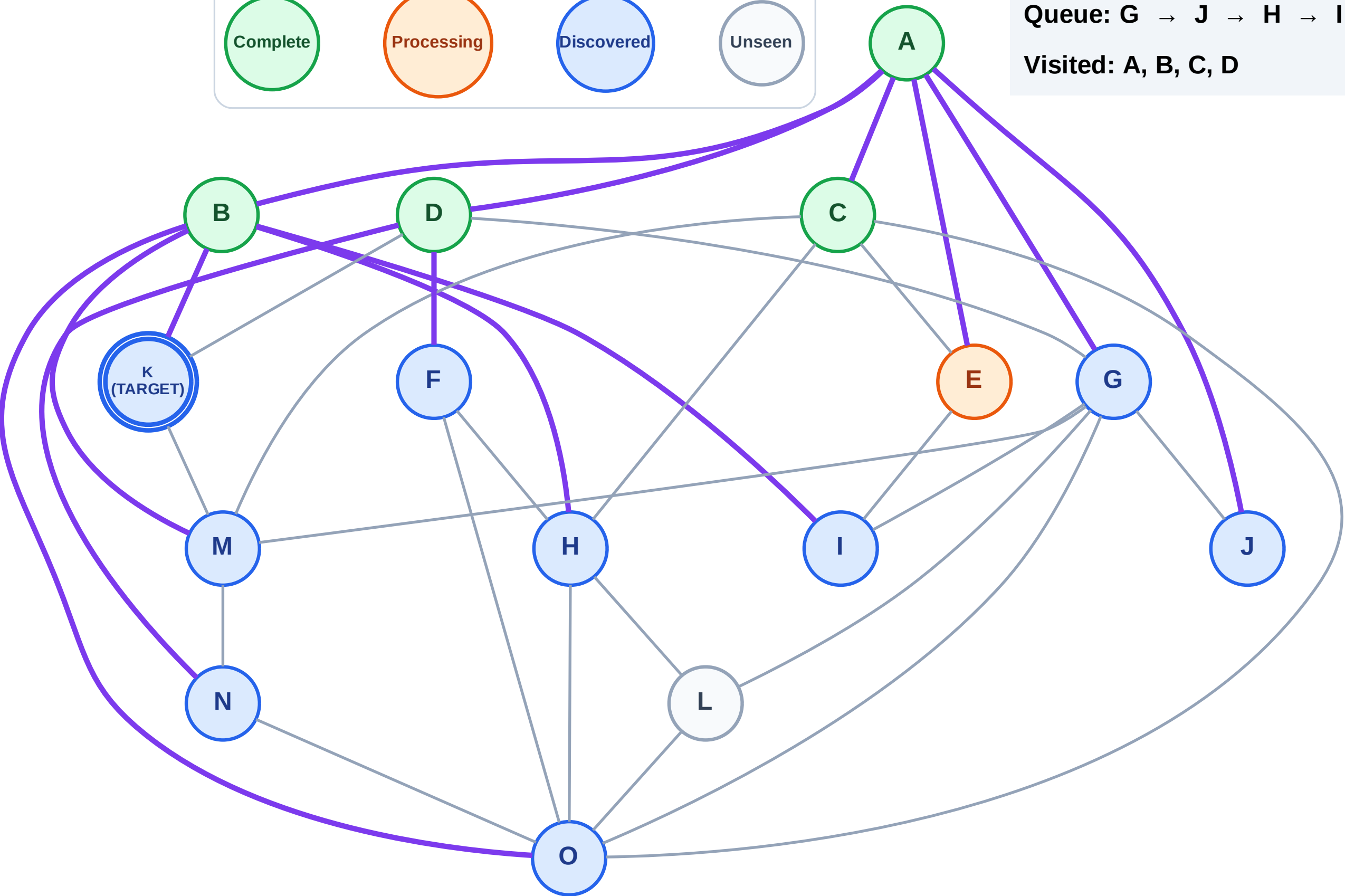
Step 10: Process node E

Legend



Queue: G → J → H → I → K → M → O → F → N

Visited: A, B, C, D



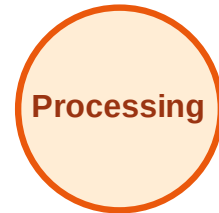
BFS Traversal

Step 11: No new neighbors from E

Legend



Complete



Processing



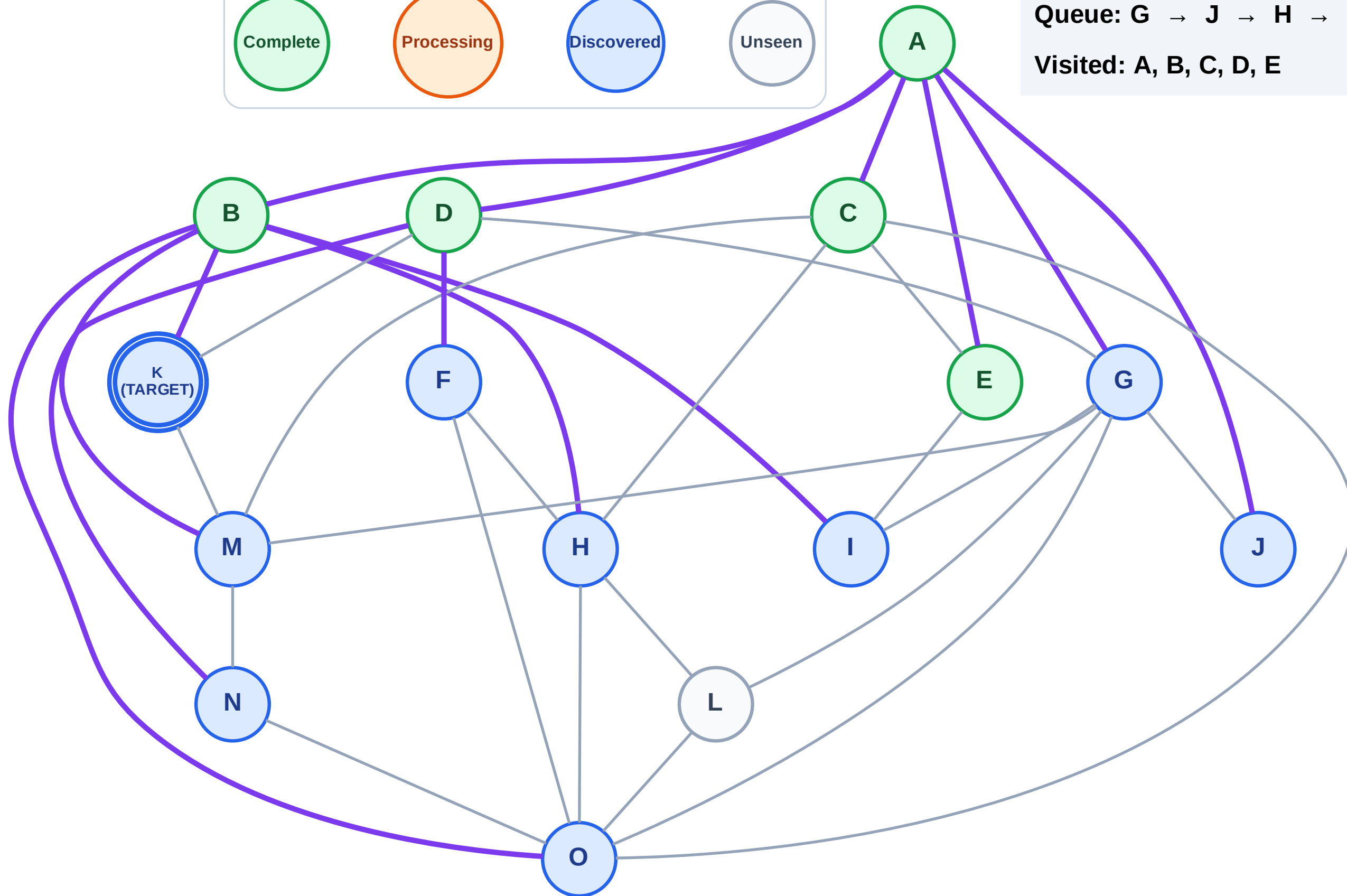
Discovered



Unseen

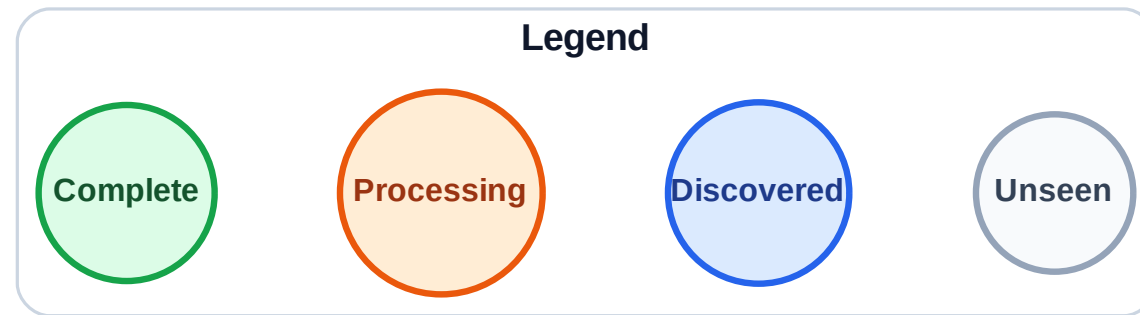
Queue: G → J → H → I → K → M → O → F → N

Visited: A, B, C, D, E



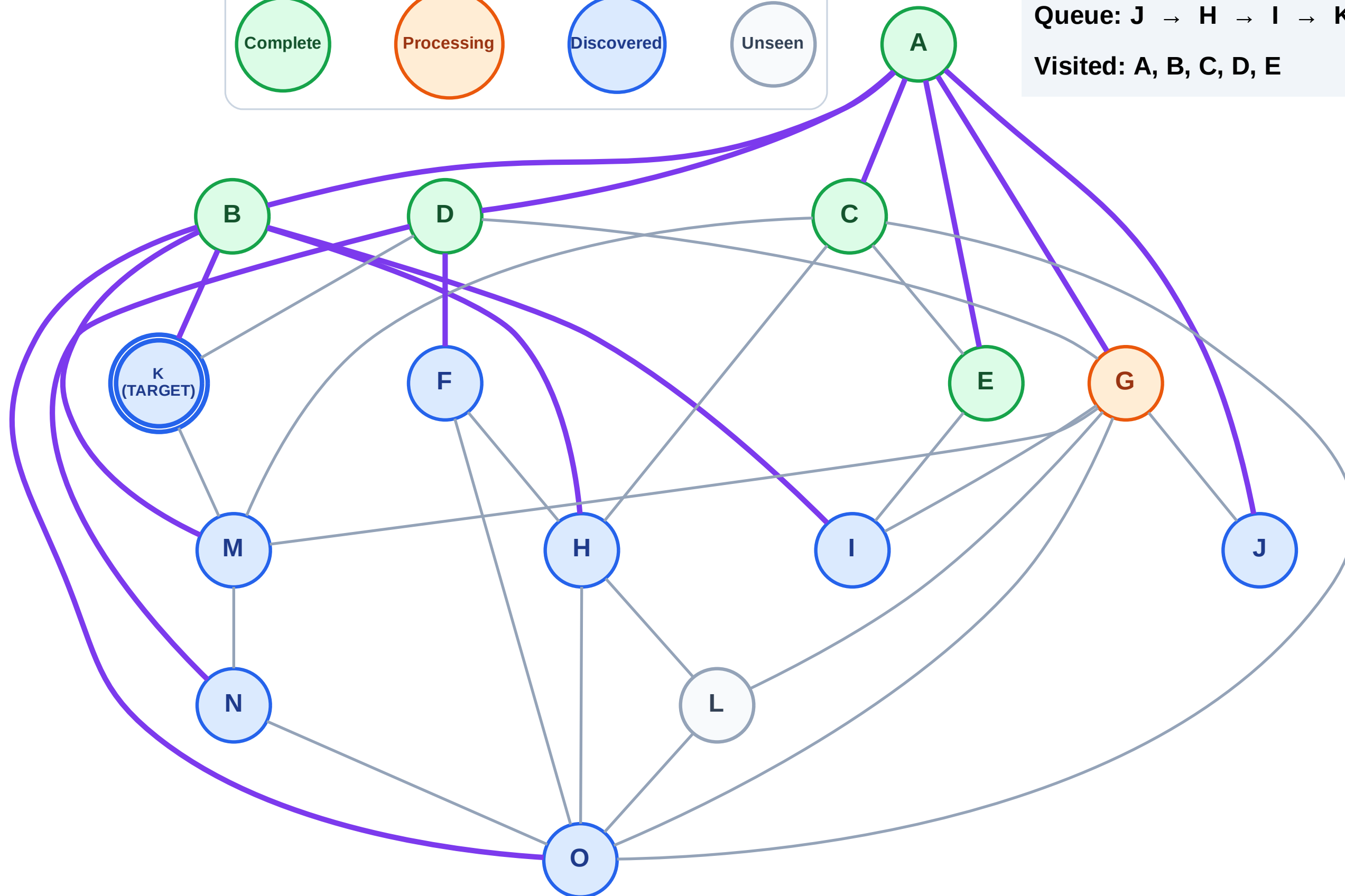
BFS Traversal

Step 12: Process node G



Queue: J → H → I → K → M → O → F → N

Visited: A, B, C, D, E



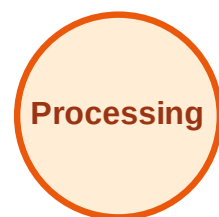
BFS Traversal

Step 13: Discover L from G

Legend



Complete



Processing



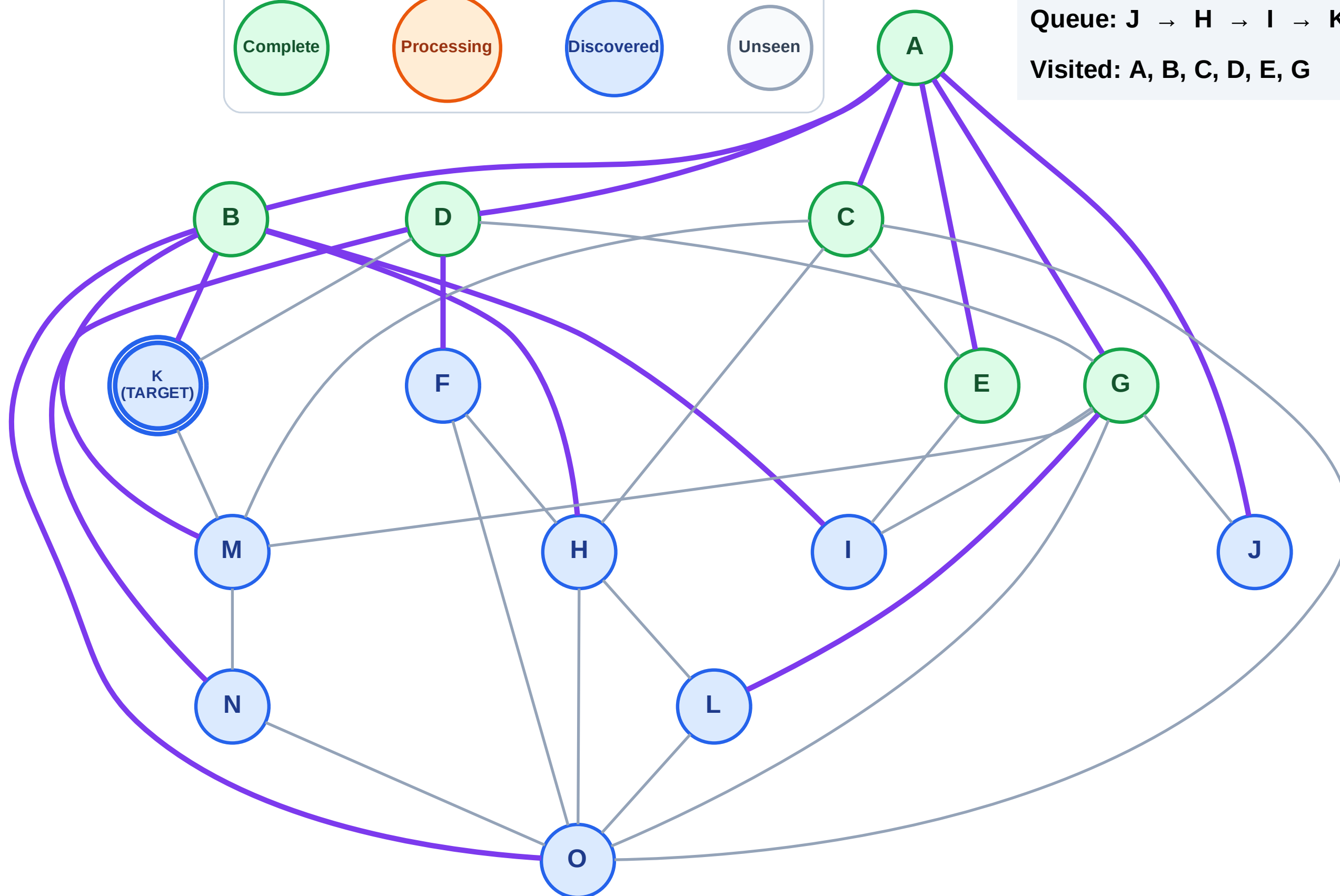
Discovered



Unseen

Queue: J → H → I → K → M → O → F → N → L

Visited: A, B, C, D, E, G



BFS Traversal

Step 14: Process node J

Legend

Complete

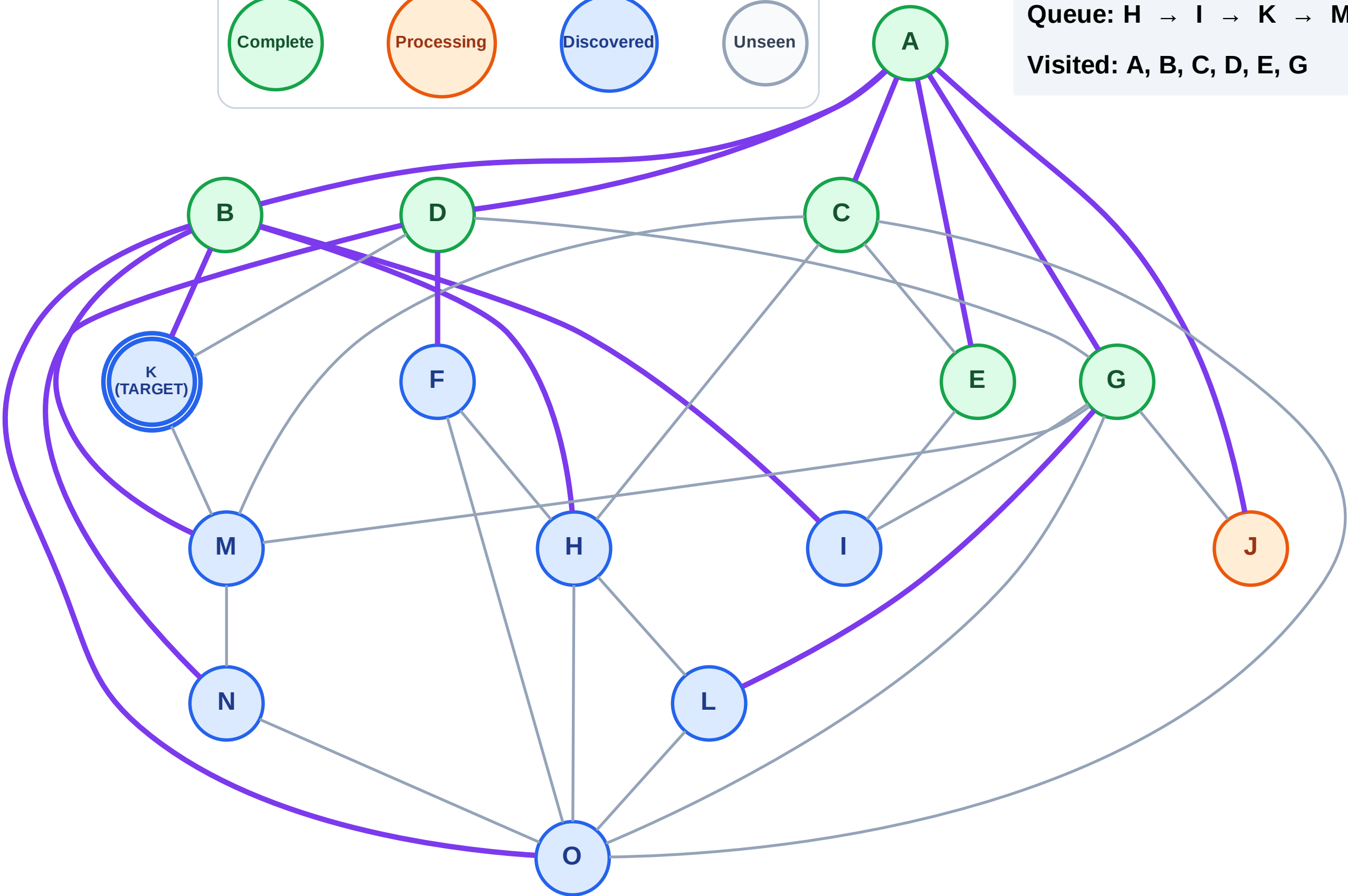
Processing

Discovered

Unseen

Queue: H → I → K → M → O → F → N → L

Visited: A, B, C, D, E, G



BFS Traversal

Step 15: No new neighbors from J

Legend

Complete

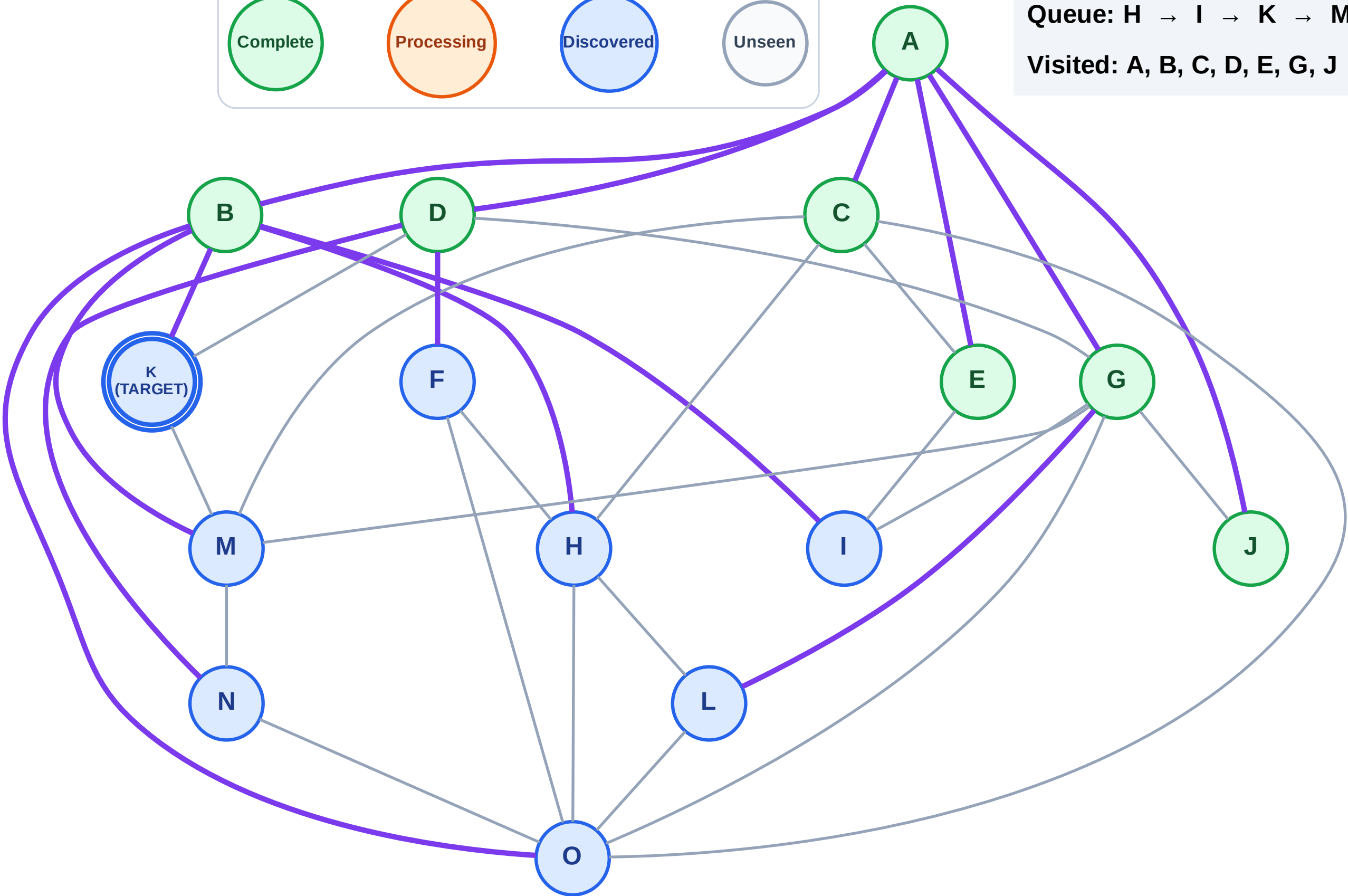
Processing

Discovered

Unseen

Queue: H → I → K → M → O → F → N → L

Visited: A, B, C, D, E, G, J



BFS Traversal

Step 16: Process node H

Legend

Complete

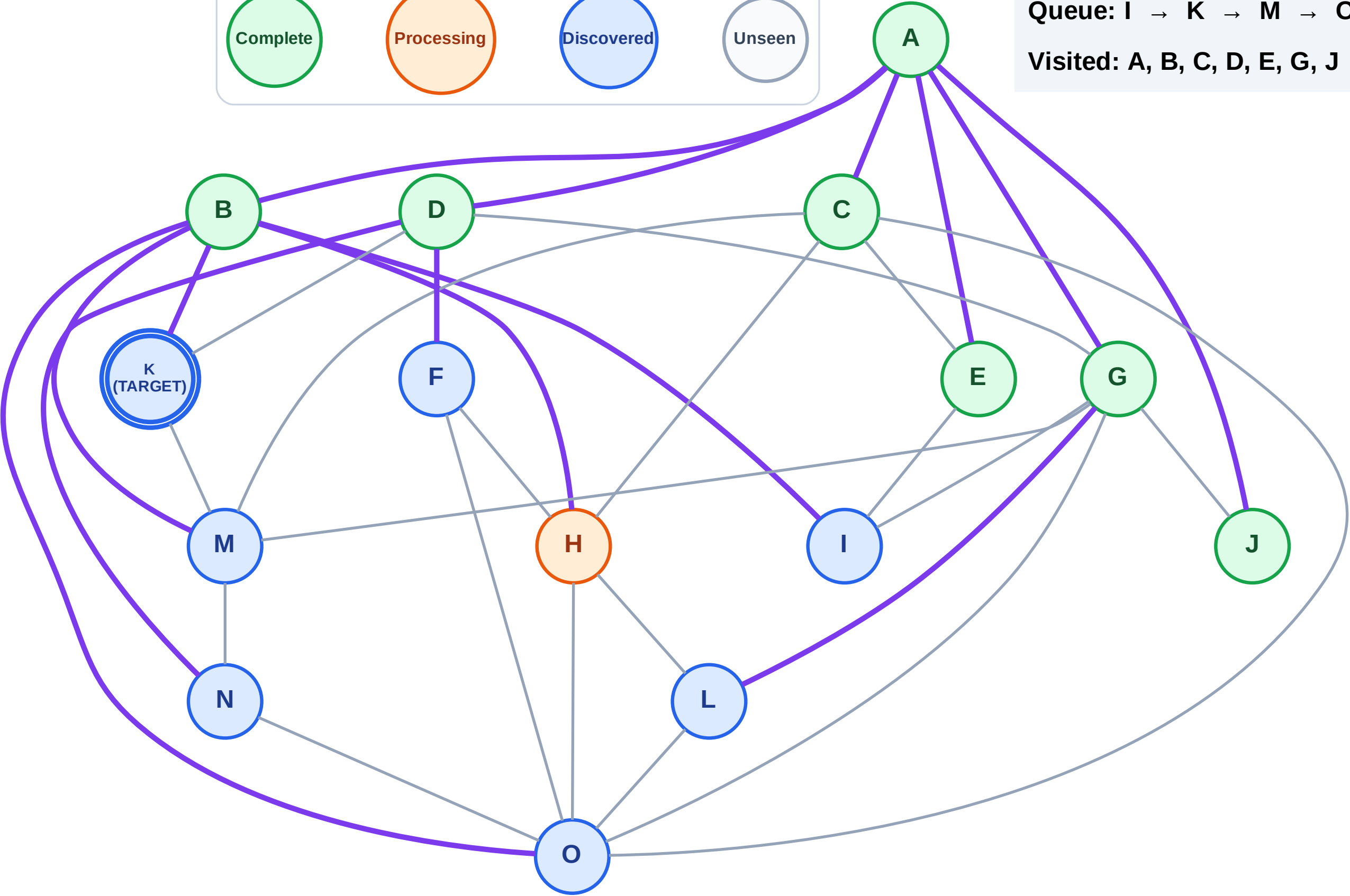
Processing

Discovered

Unseen

Queue: I → K → M → O → F → N → L

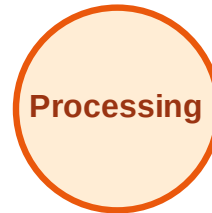
Visited: A, B, C, D, E, G, J



BFS Traversal

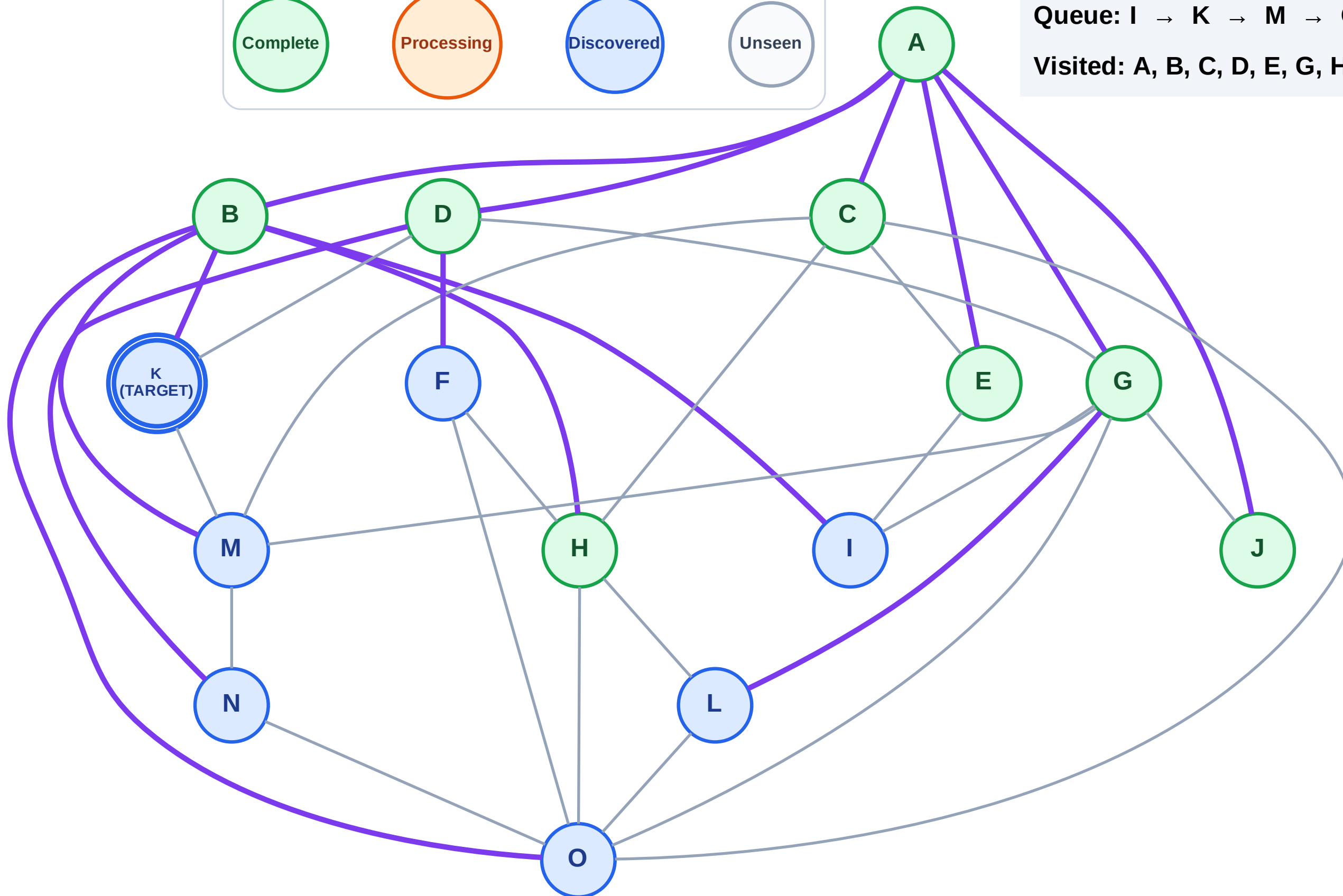
Step 17: No new neighbors from H

Legend



Queue: I → K → M → O → F → N → L

Visited: A, B, C, D, E, G, H, J



BFS Traversal

Step 18: Process node I

Legend

Complete

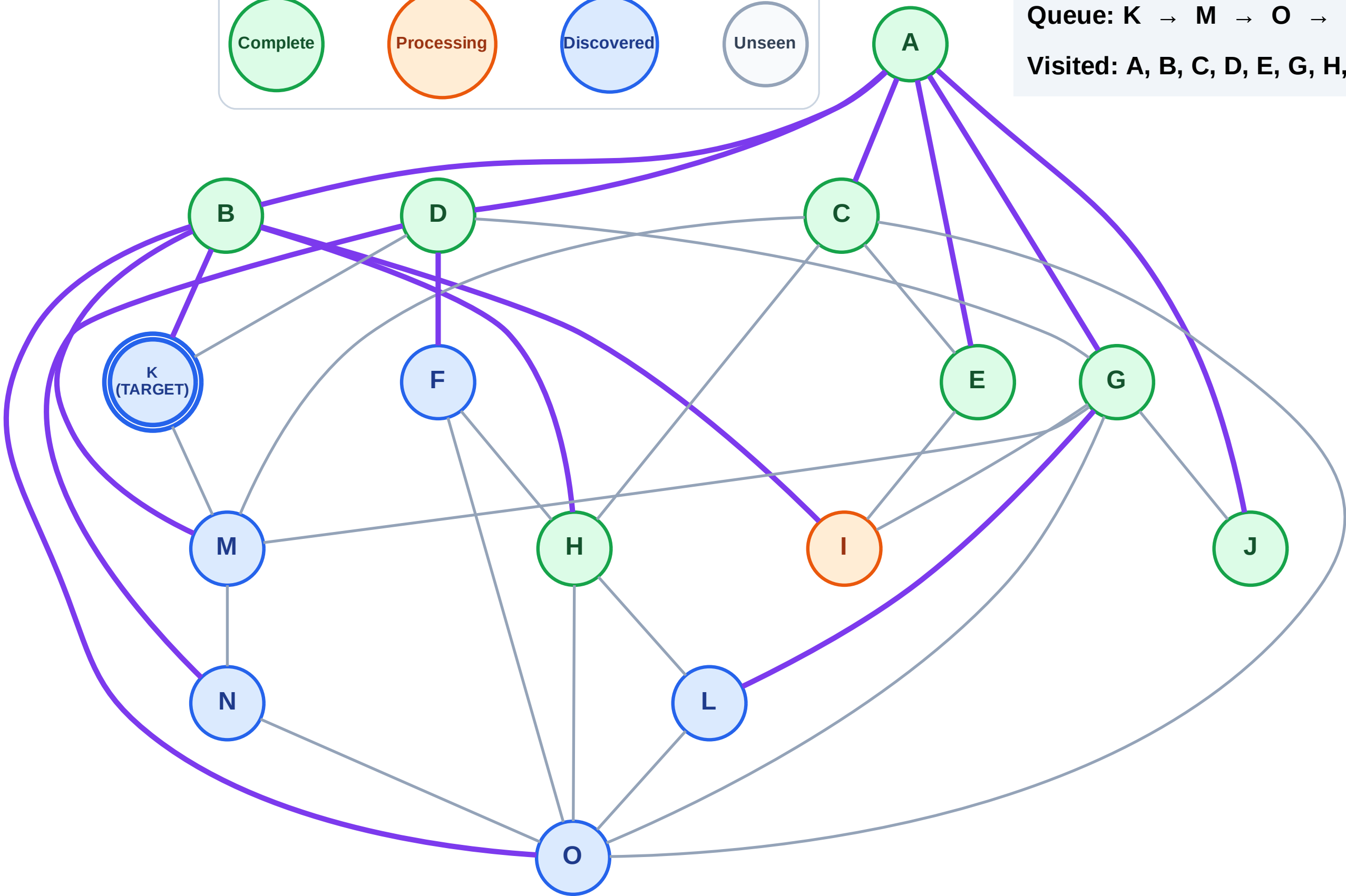
Processing

Discovered

Unseen

Queue: K → M → O → F → N → L

Visited: A, B, C, D, E, G, H, J



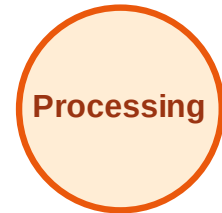
BFS Traversal

Step 19: No new neighbors from I

Legend



Complete



Processing



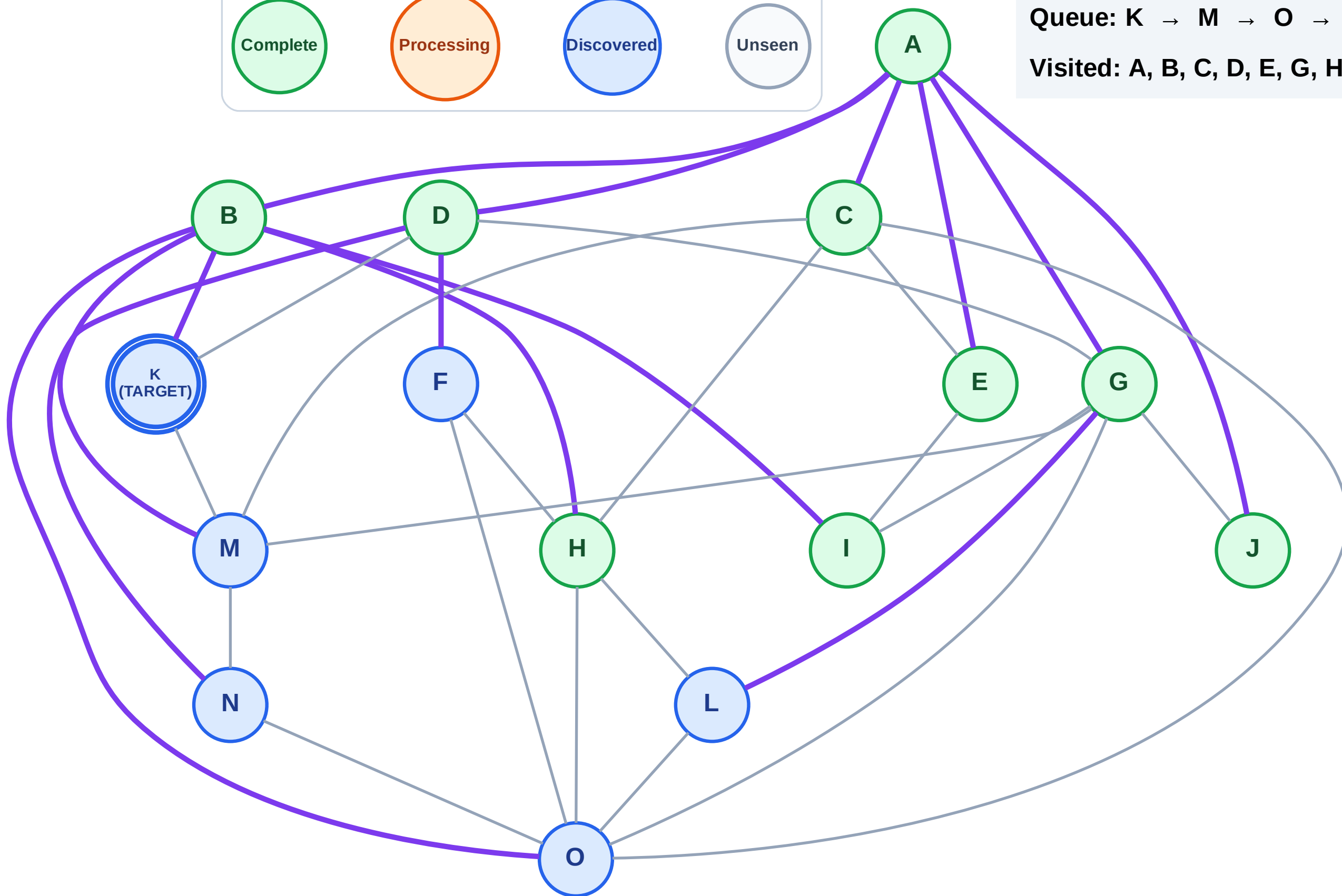
Discovered



Unseen

Queue: K → M → O → F → N → L

Visited: A, B, C, D, E, G, H, I, J



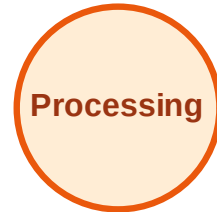
BFS Traversal

Step 20: Process node K

Legend



Complete



Processing



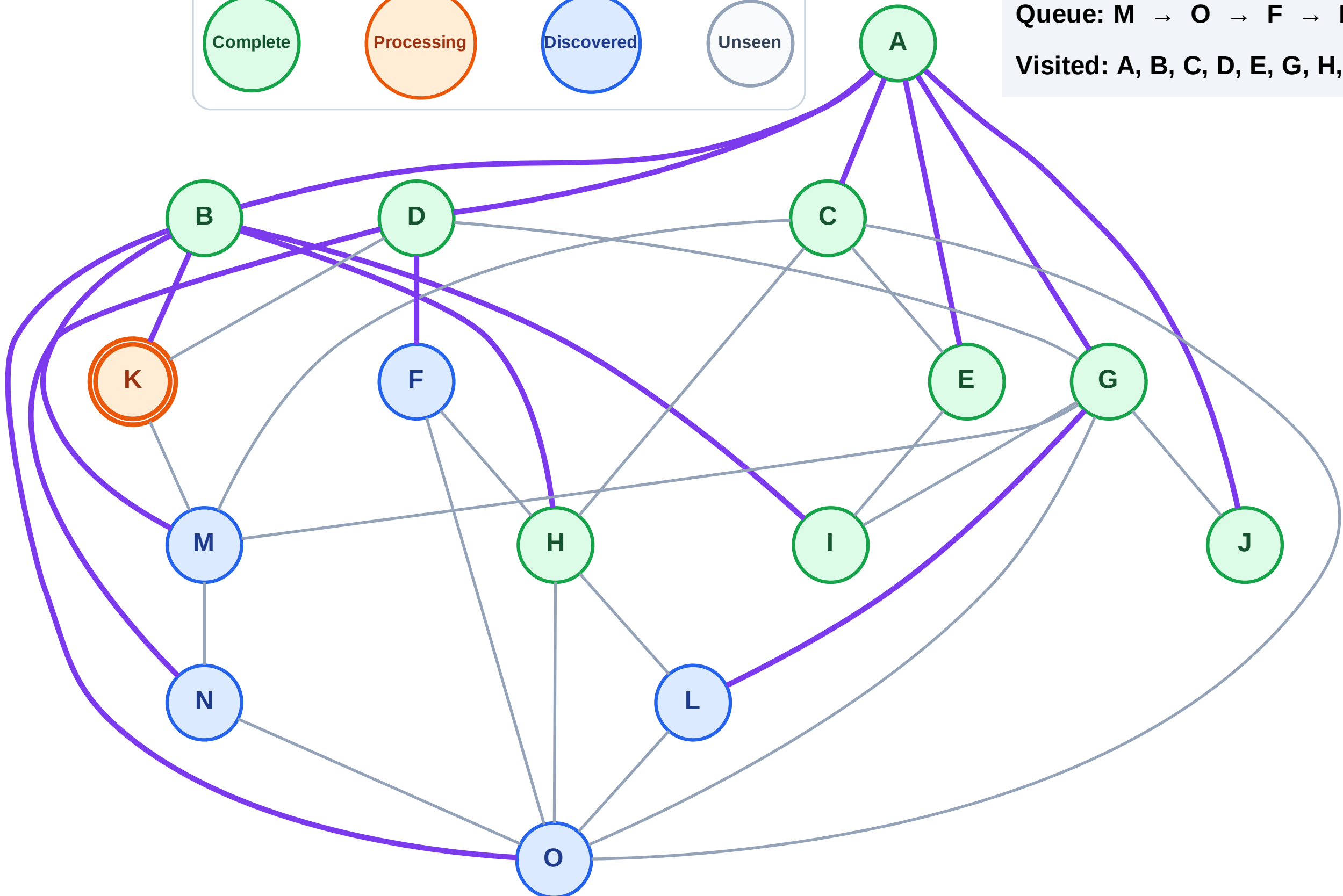
Discovered



Unseen

Queue: M → O → F → N → L

Visited: A, B, C, D, E, G, H, I, J



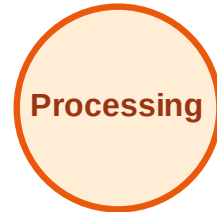
BFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Complete



Processing



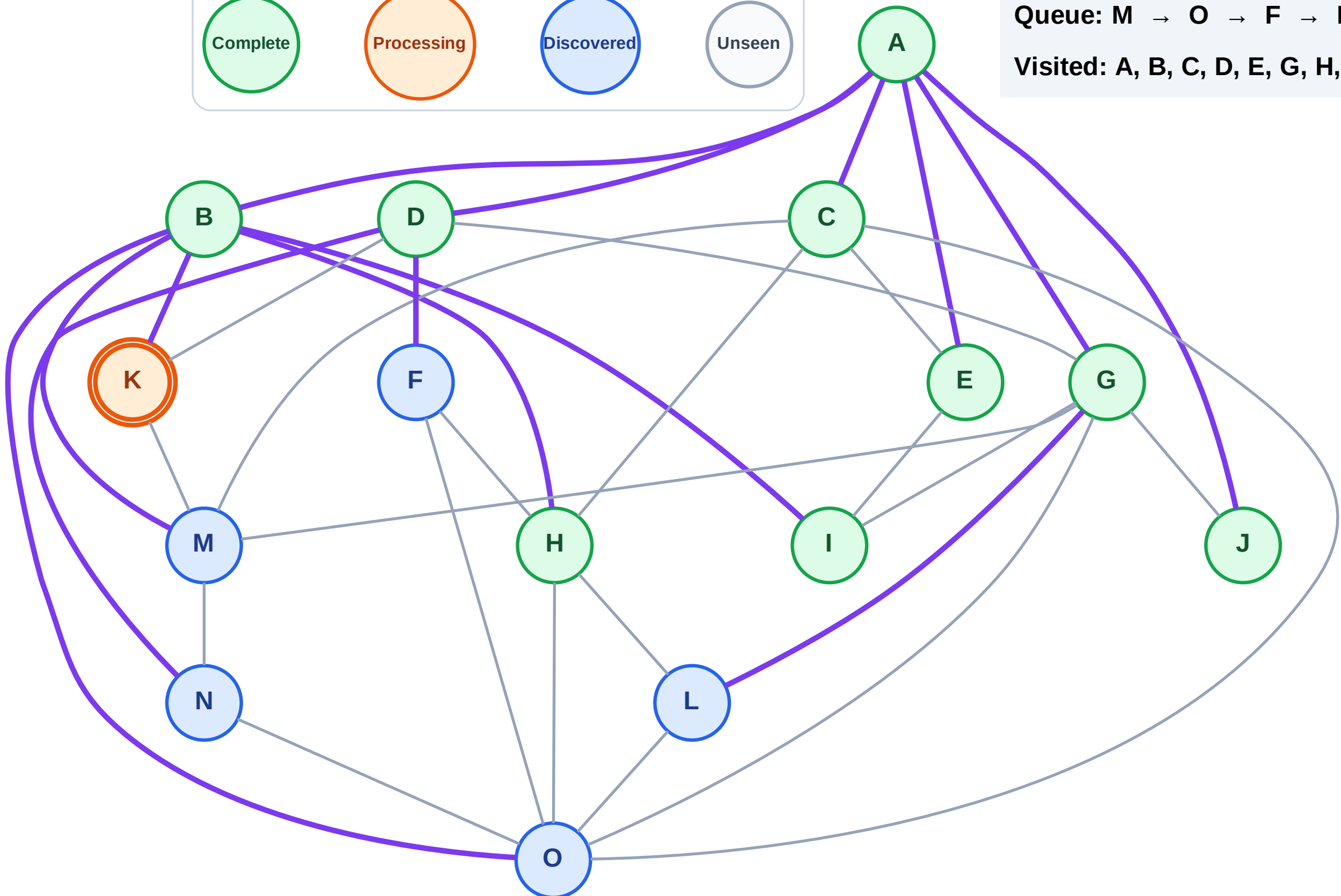
Discovered



Unseen

Queue: M → O → F → N → L

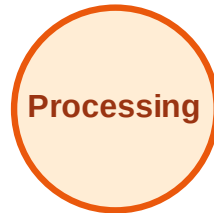
Visited: A, B, C, D, E, G, H, I, J, K



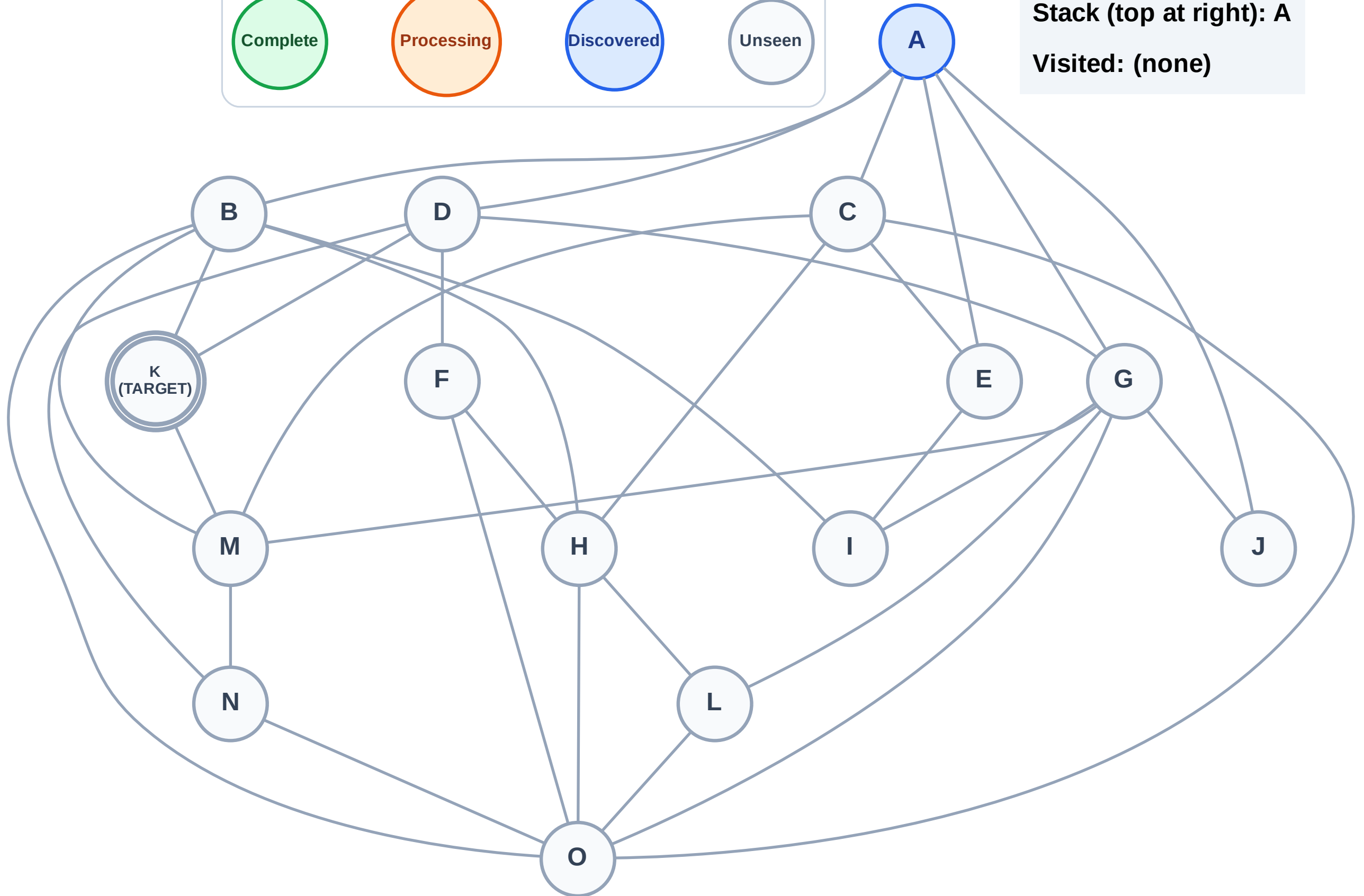
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A
Visited: (none)



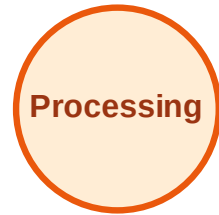
DFS Traversal

Step 2: Process node A

Legend



Complete



Processing



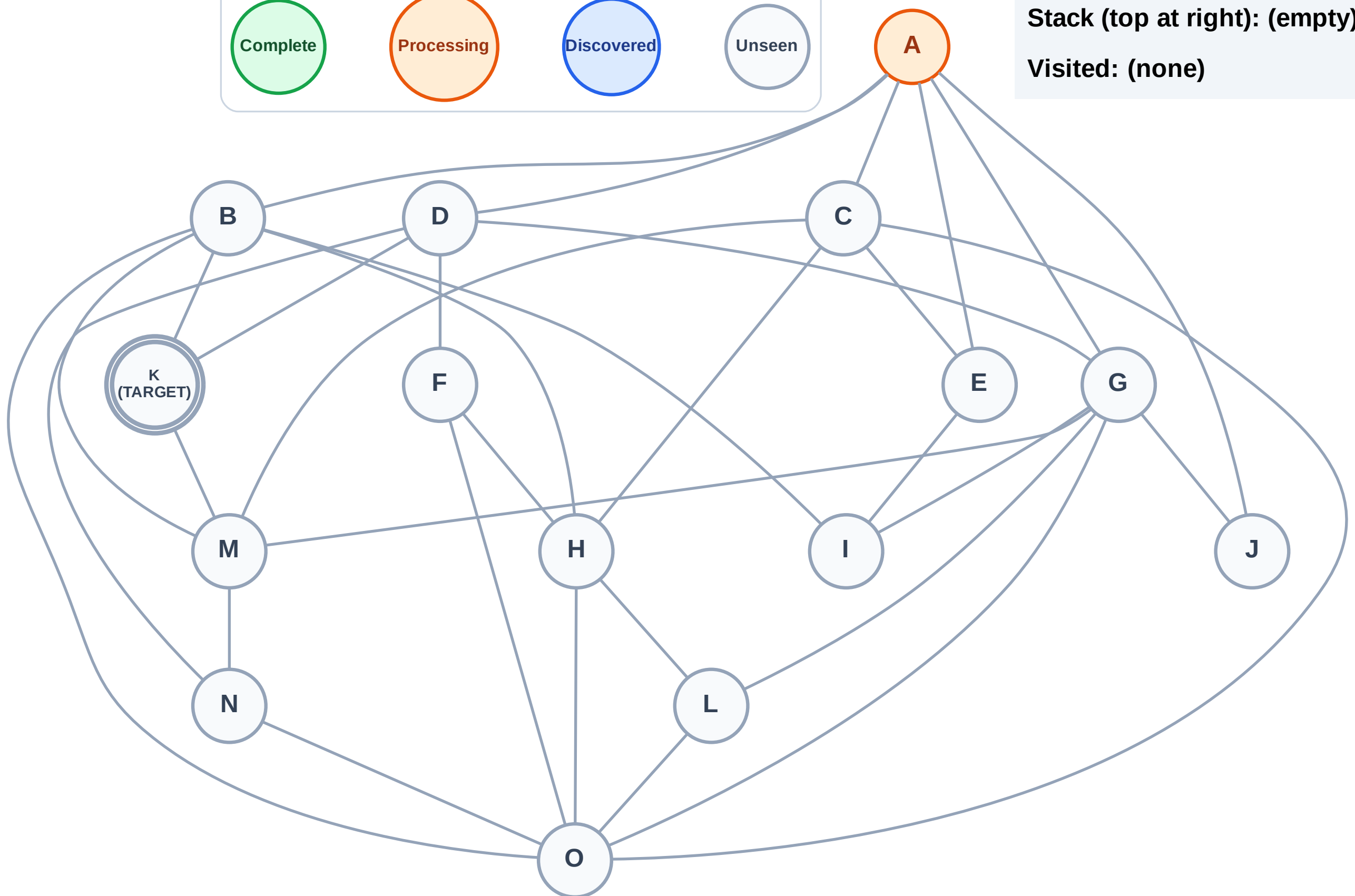
Discovered



Unseen

Stack (top at right): (empty)

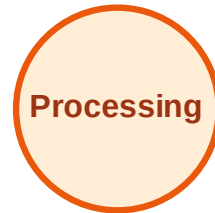
Visited: (none)



DFS Traversal

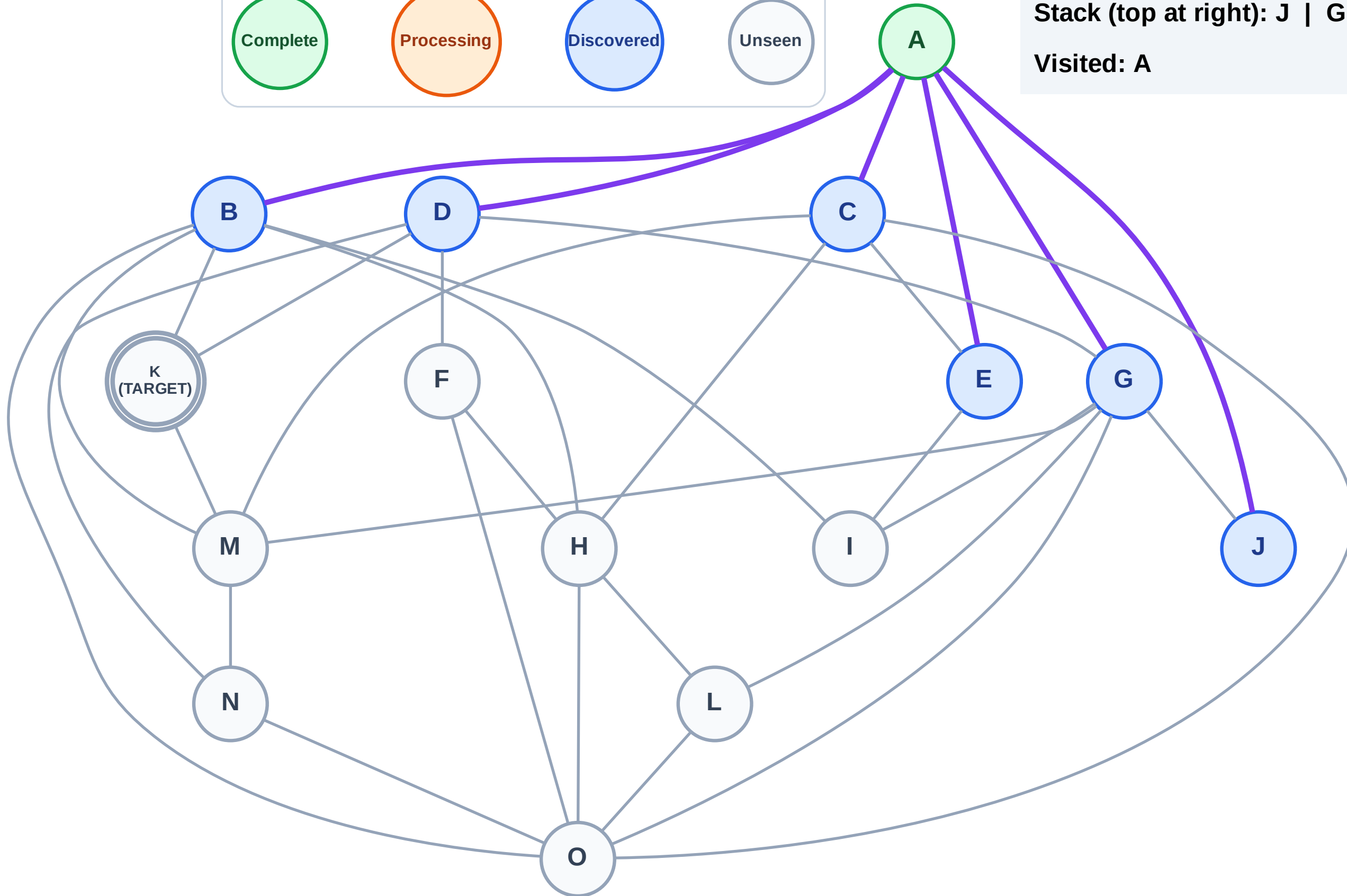
Step 3: Discover J, G, E, D, C, B from A

Legend



Stack (top at right): J | G | E | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

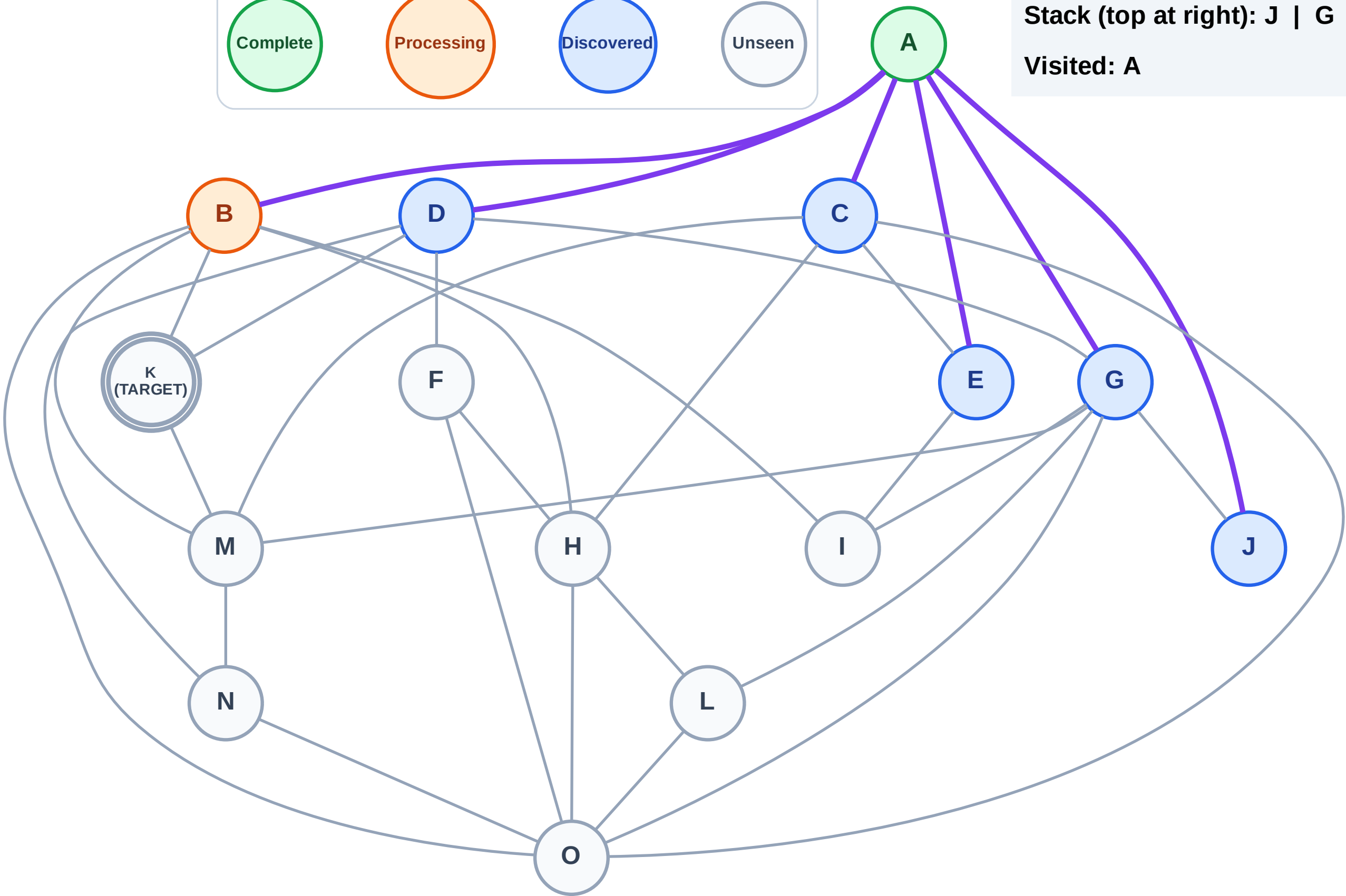
Complete

Processing

Discovered

Unseen

Stack (top at right): J | G | E | D | C
Visited: A



DFS Traversal

Step 5: Discover O, M, K, I, H from B

Legend

Complete

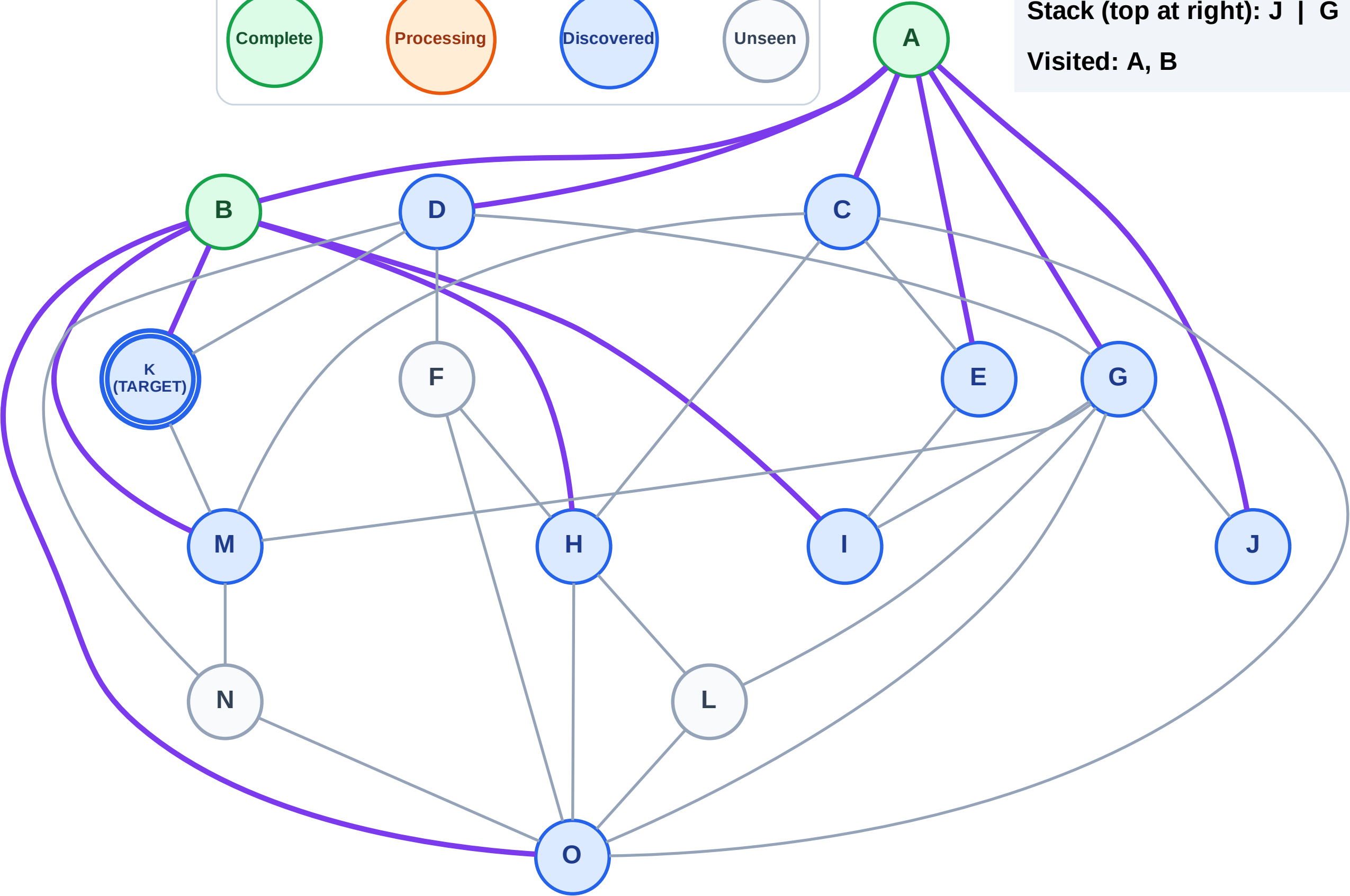
Processing

Discovered

Unseen

Stack (top at right): J | G | E | D | C | O | M | K | I | H

Visited: A, B



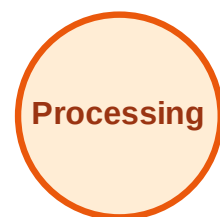
DFS Traversal

Step 6: Process node H

Legend



Complete



Processing



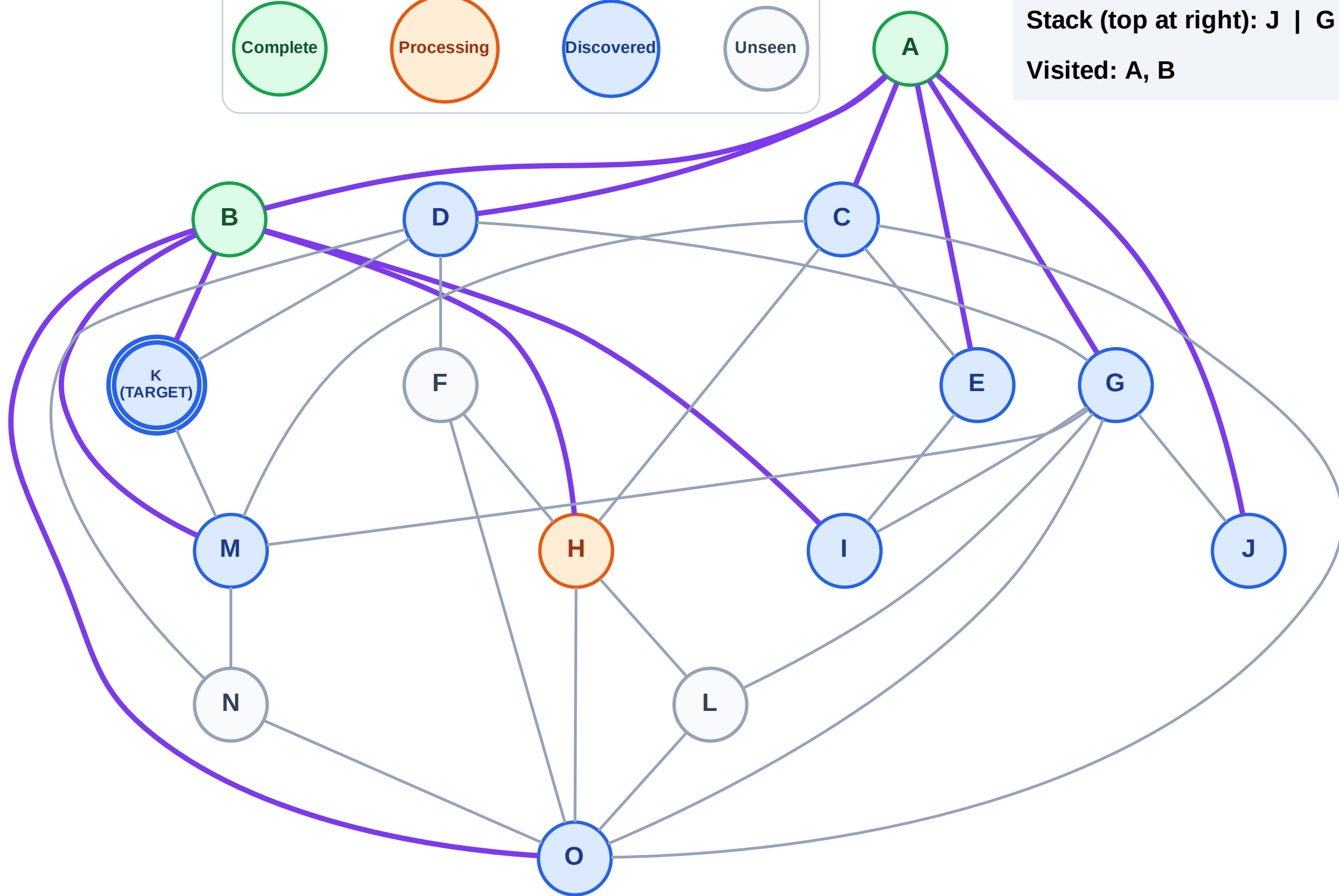
Discovered



Unseen

Stack (top at right): J | G | E | D | C | O | M | K | I

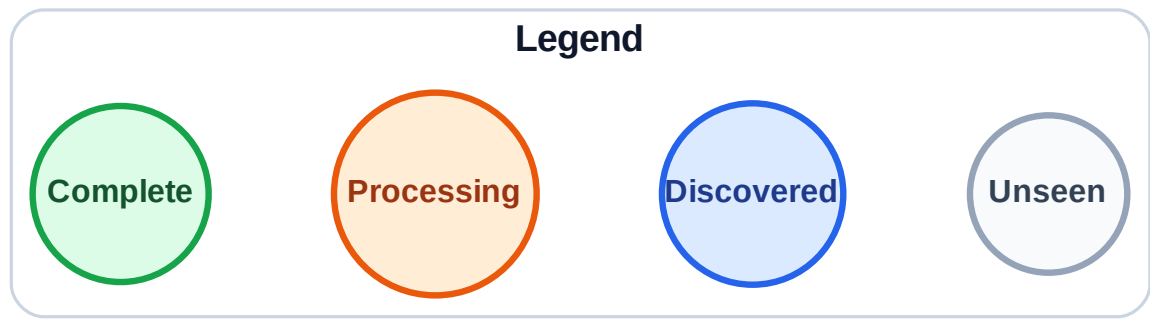
Visited: A, B



Step 7: Discover L, F from H

Stack (top at right): J | G | E | D | C | O | M | K | I | L | F

Visited: A, B, H



DFS Traversal

Step 8: Process node F

Legend

Complete

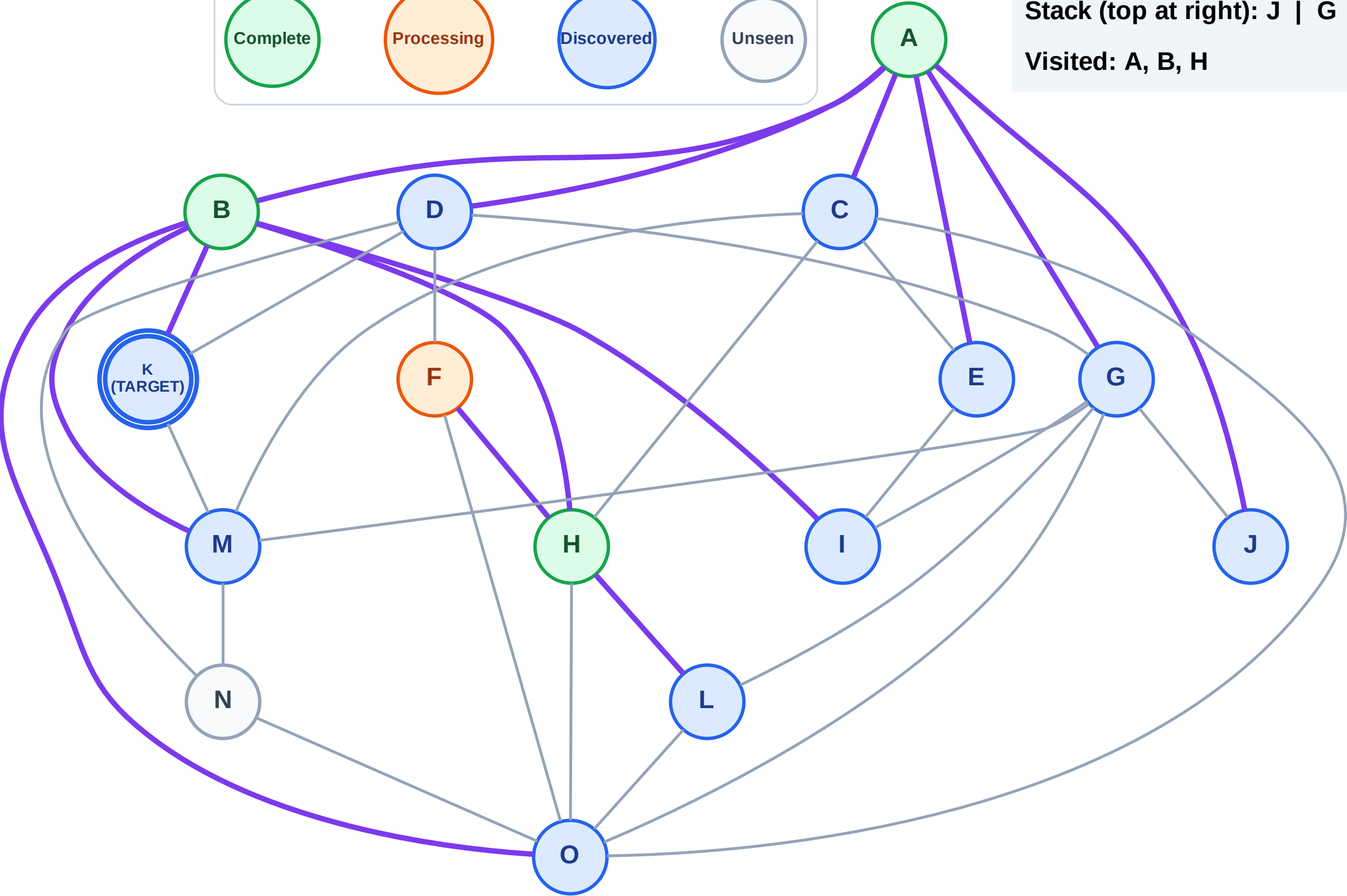
Processing

Discovered

Unseen

Stack (top at right): J | G | E | D | C | O | M | K | I | L

Visited: A, B, H



Step 9: No new neighbors from F

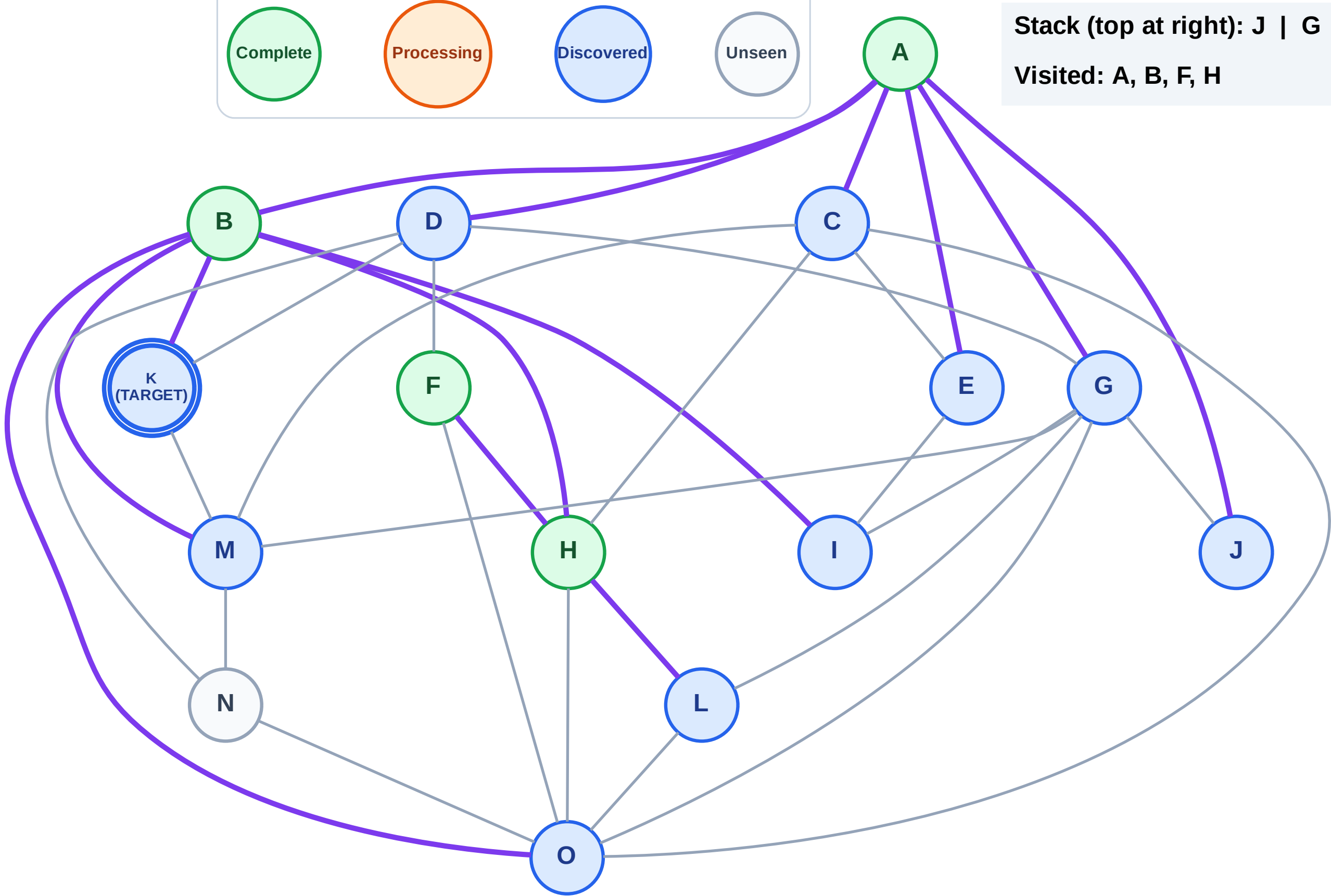
Legend

Processing

Discovered

Unseen

Visited: A, B, F, H



DFS Traversal

Step 10: Process node L

Legend

Complete

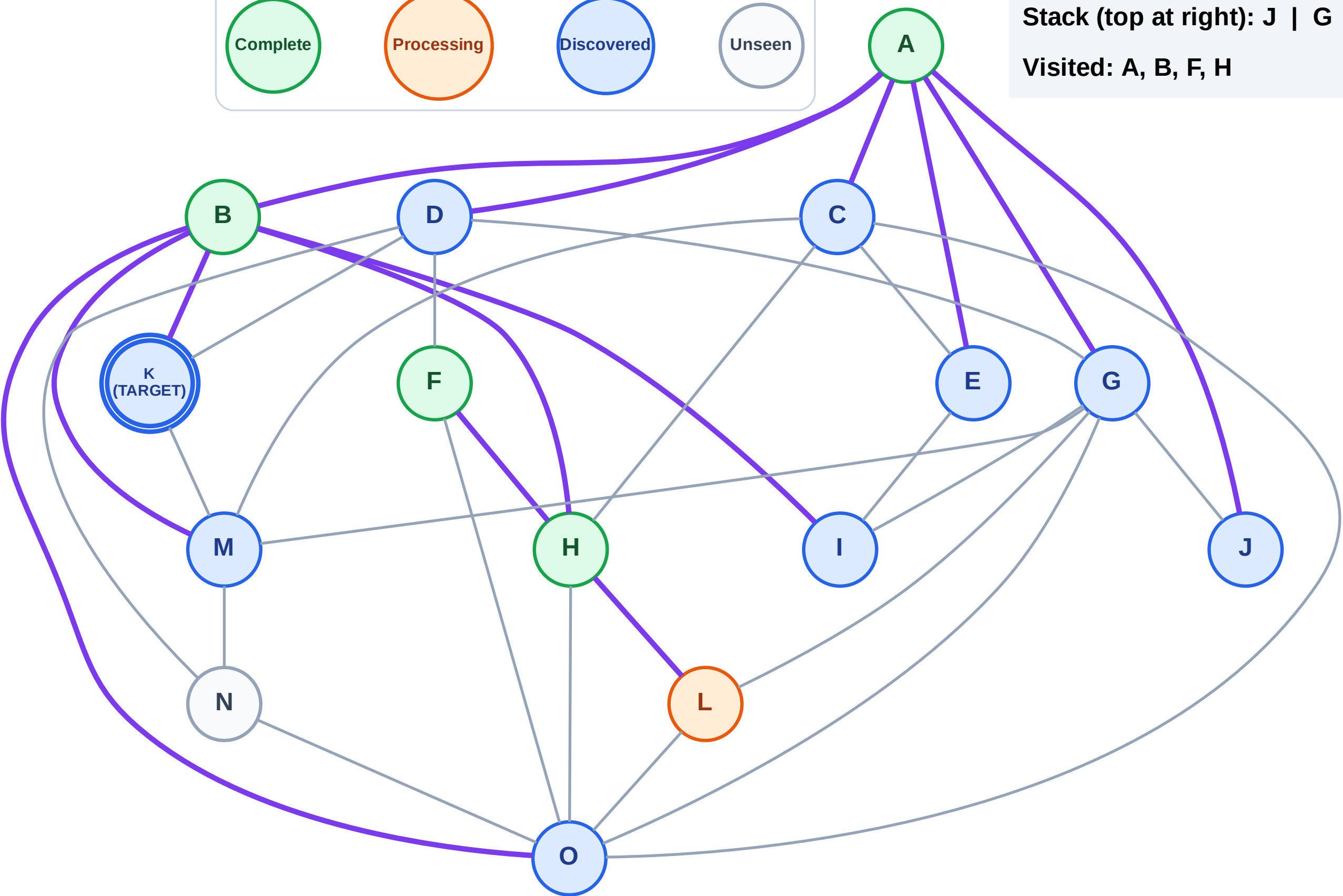
Processing

Discovered

Unseen

Stack (top at right): J | G | E | D | C | O | M | K | I

Visited: A, B, F, H



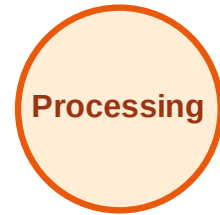
DFS Traversal

Step 11: No new neighbors from L

Legend



Complete



Processing



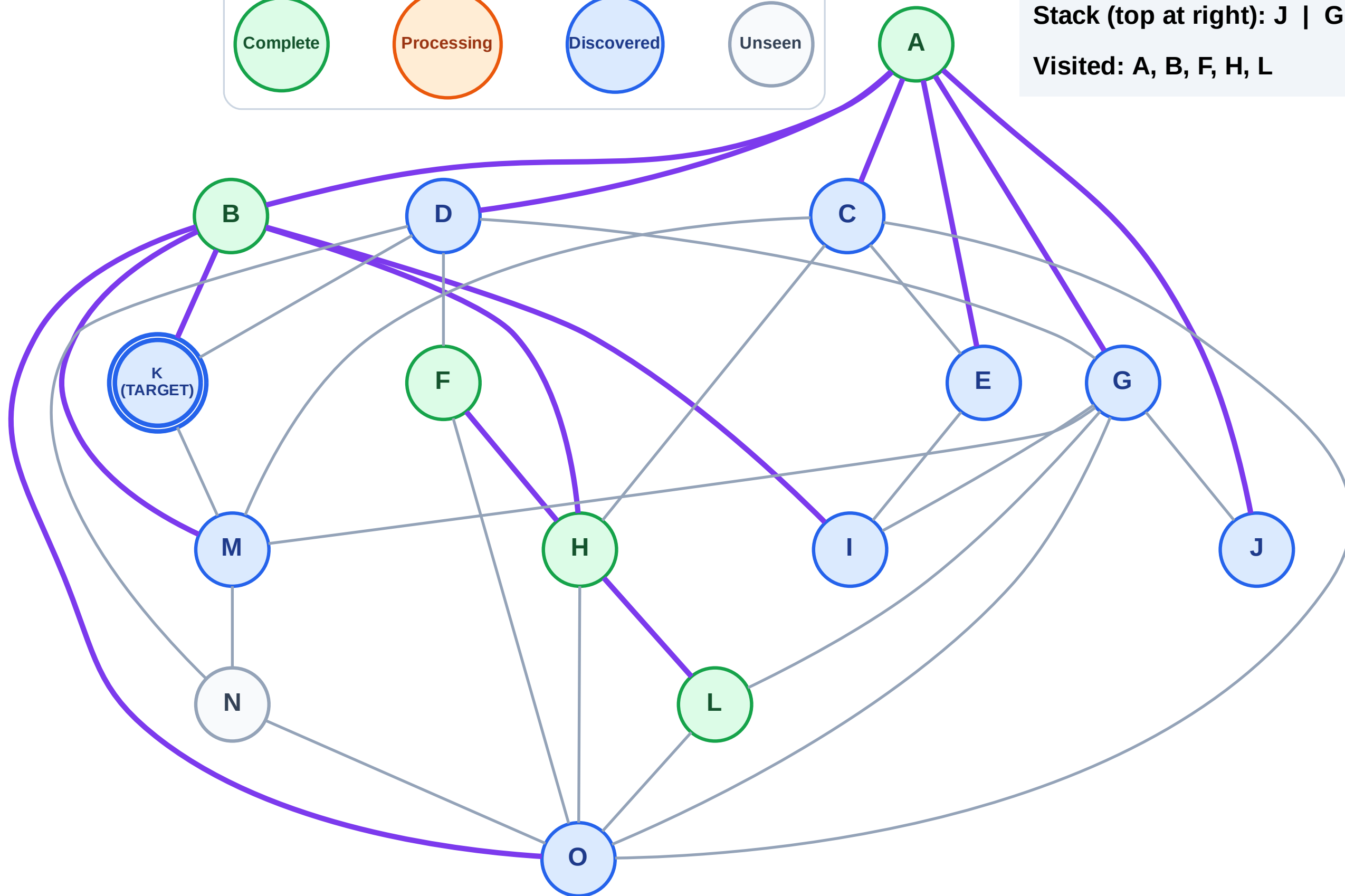
Discovered



Unseen

Stack (top at right): J | G | E | D | C | O | M | K | I

Visited: A, B, F, H, L



DFS Traversal

Step 12: Process node I

Legend

Complete

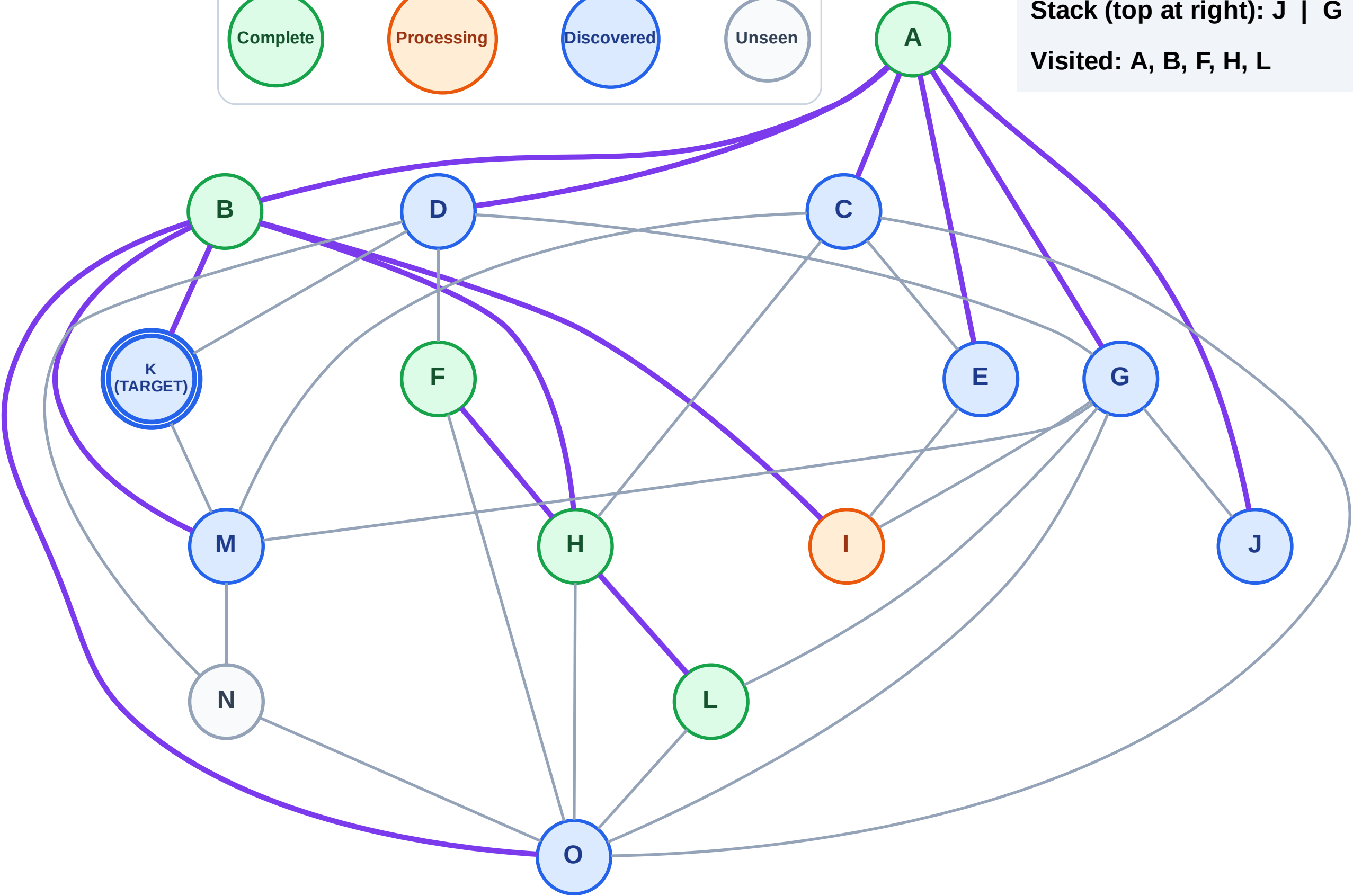
Processing

Discovered

Unseen

Stack (top at right): J | G | E | D | C | O | M | K

Visited: A, B, F, H, L



DFS Traversal

Step 13: No new neighbors from I

Legend

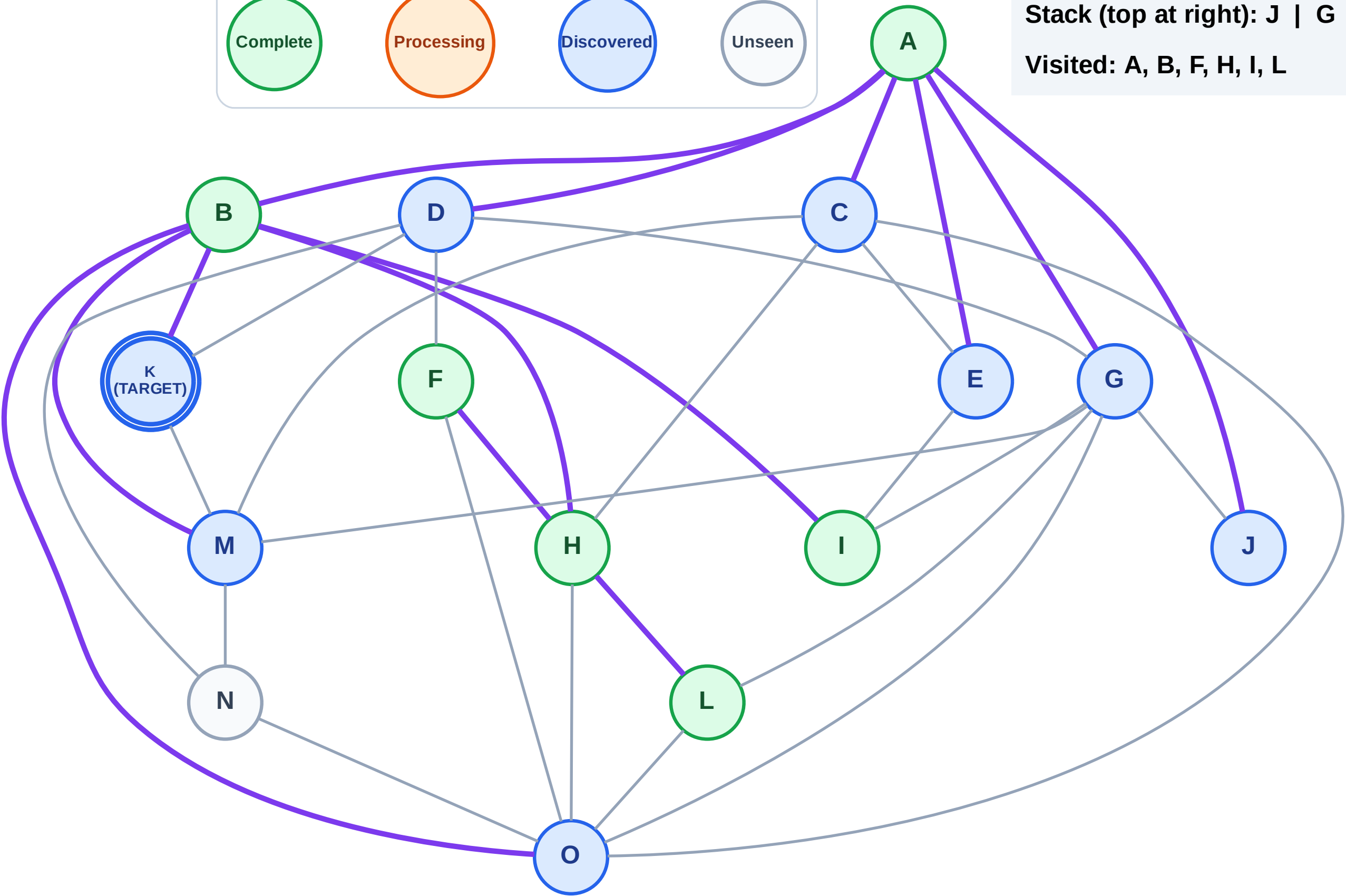
Complete

Processing

Discovered

Unseen

Stack (top at right): J | G | E | D | C | O | M | K
Visited: A, B, F, H, I, L



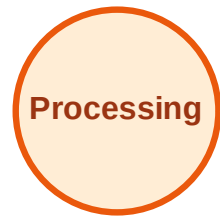
DFS Traversal

Step 14: Process node K

Legend



Complete



Processing



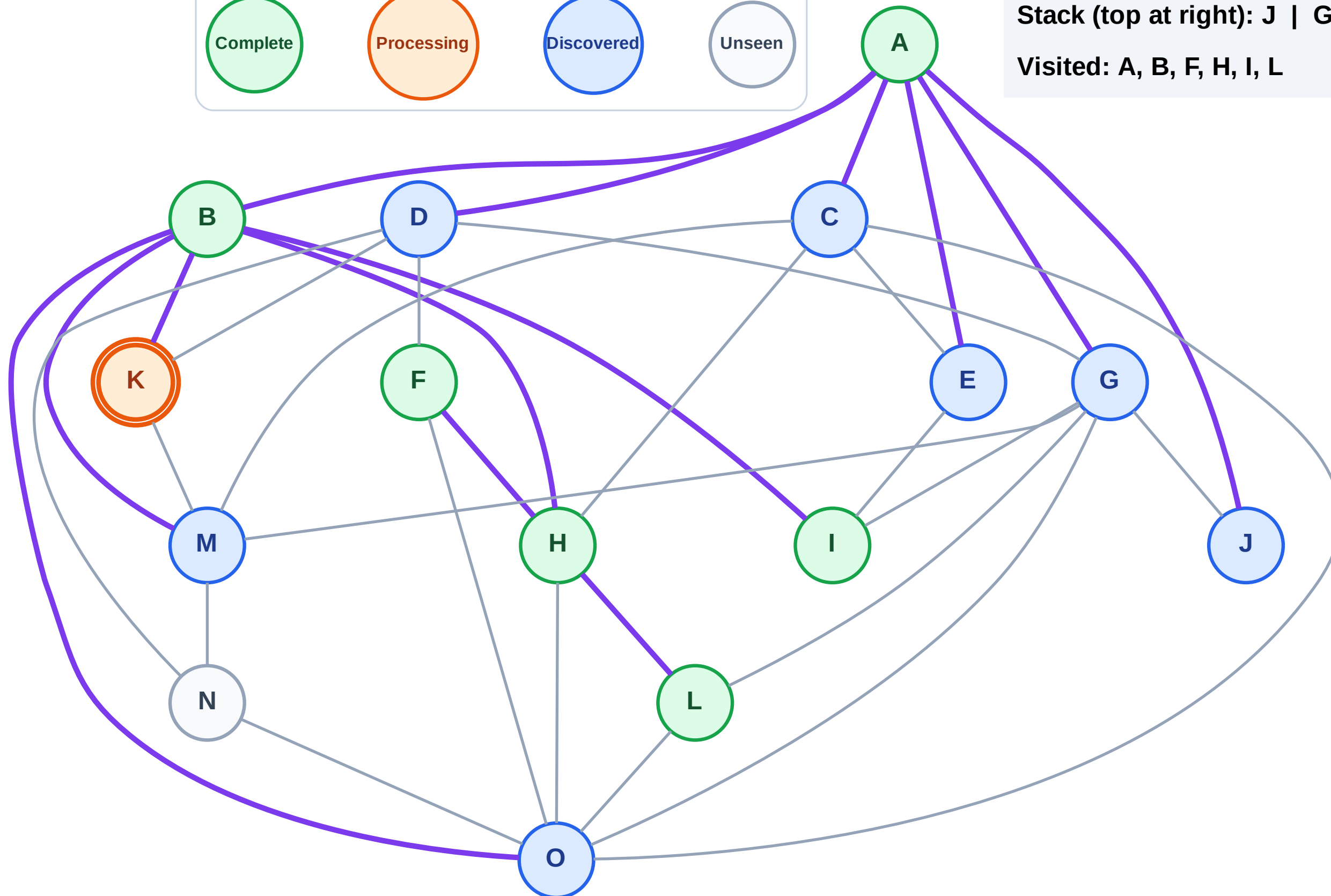
Discovered



Unseen

Stack (top at right): J | G | E | D | C | O | M

Visited: A, B, F, H, I, L



DFS Traversal

Step 15: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): J | G | E | D | C | O | M

Visited: A, B, F, H, I, K, L

